

Digital Image Processing 8:

Morphological Image Processing

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Contents

Once segmentation is complete, morphological operations can be used to remove imperfections in the segmented image and provide information on the form and structure of the image. What is morphology?

- Simple morphological operations
- Compound operations
- Morphological algorithms

What Is Morphology?

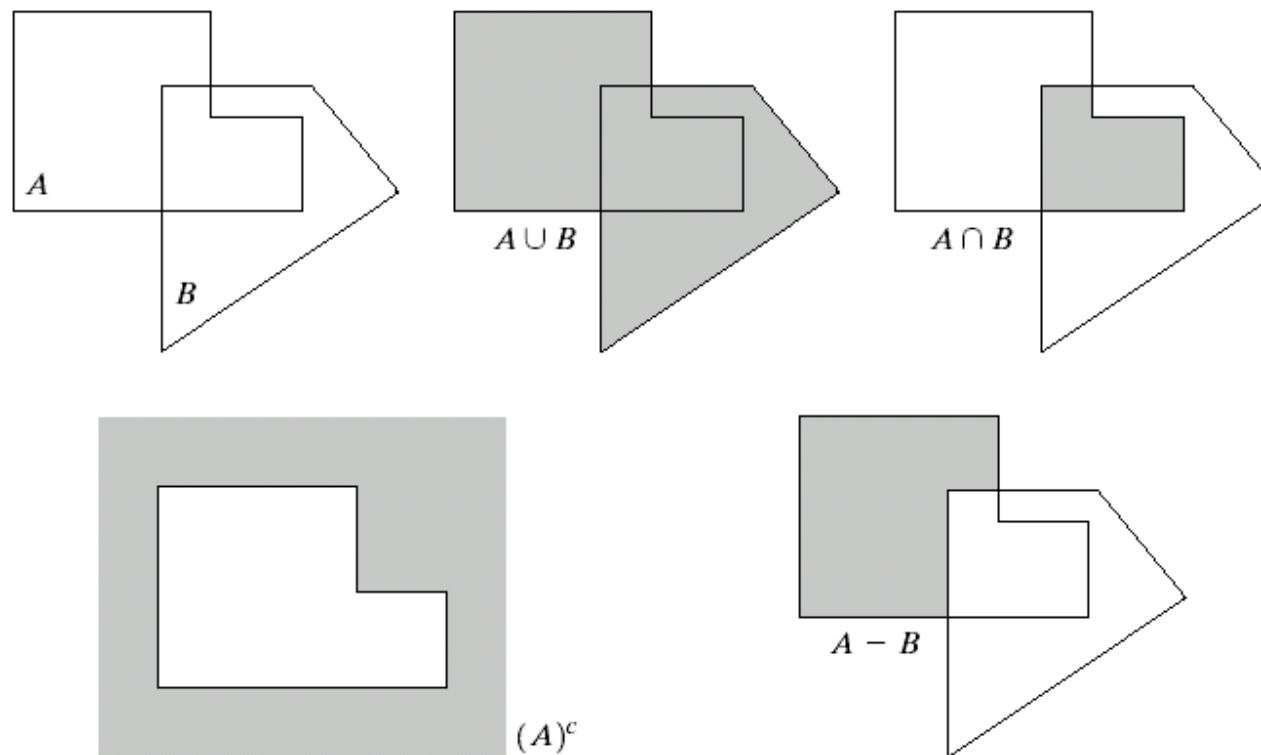
Morphological image processing (or *morphology*) describes a range of image processing techniques that deal with the **shape (or morphology) of features** in an image.

Morphological operations are typically applied to **remove imperfections introduced during segmentation**, and so typically operate on bi-level images

Mathematical framework used for:

- pre-processing
 - noise filtering, shape simplification, ...
- enhancing object structure
 - skeletonization, convex hull...
- Segmentation
 - watershed,...
- quantitative description
 - area, perimeter, ...

Basic Set Theory



a	b	c
d	e	

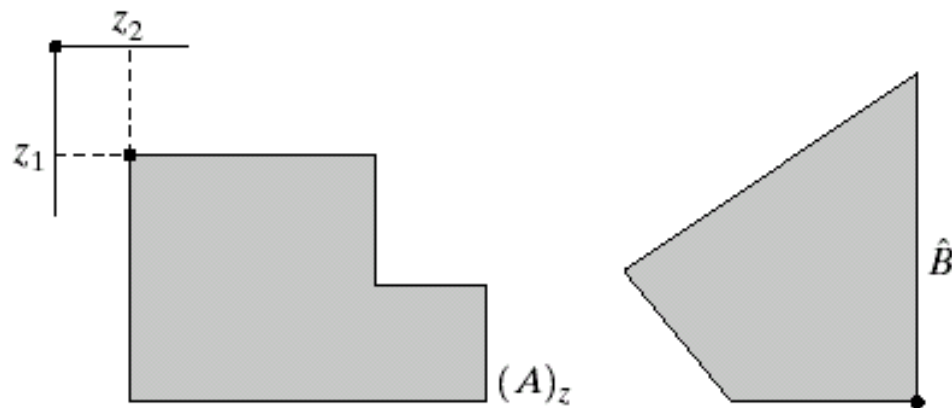
FIGURE 9.1

(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .

Reflection and Translation

$$\hat{B} = \{w \mid w \in -b, \text{ for } b \in B\}$$

$$(A)_z = \{c \mid c \in a + z, \text{ for } a \in A\}$$

**a b****FIGURE 9.2**

(a) Translation of A by z .

(b) Reflection of B . The sets A and B are from Fig. 9.1.

Logic Operations

p	q	p AND q (also $p \cdot q$)	p OR q (also $p + q$)	NOT (p) (also $\bar{p}$)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

Example

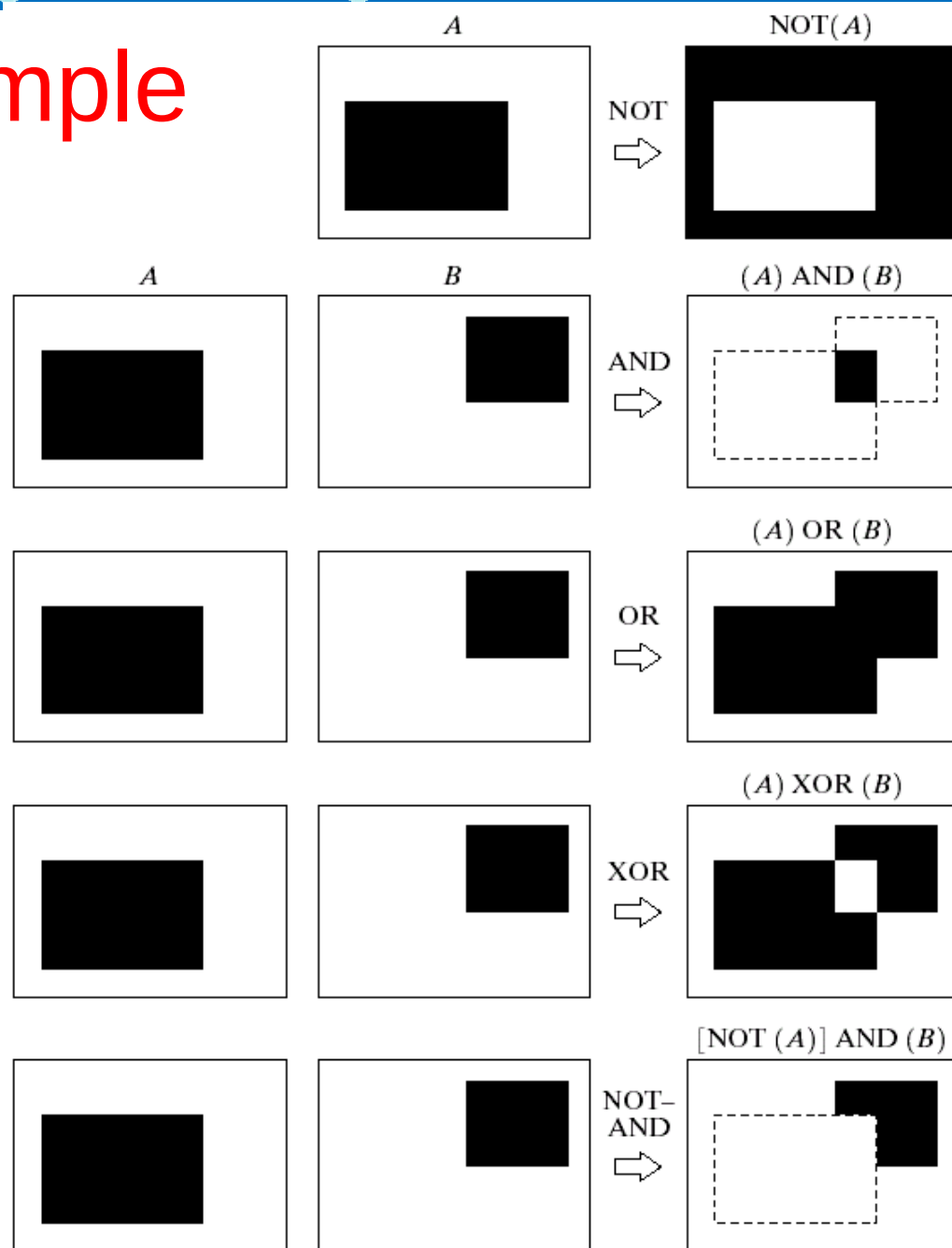
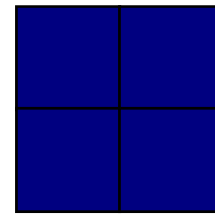
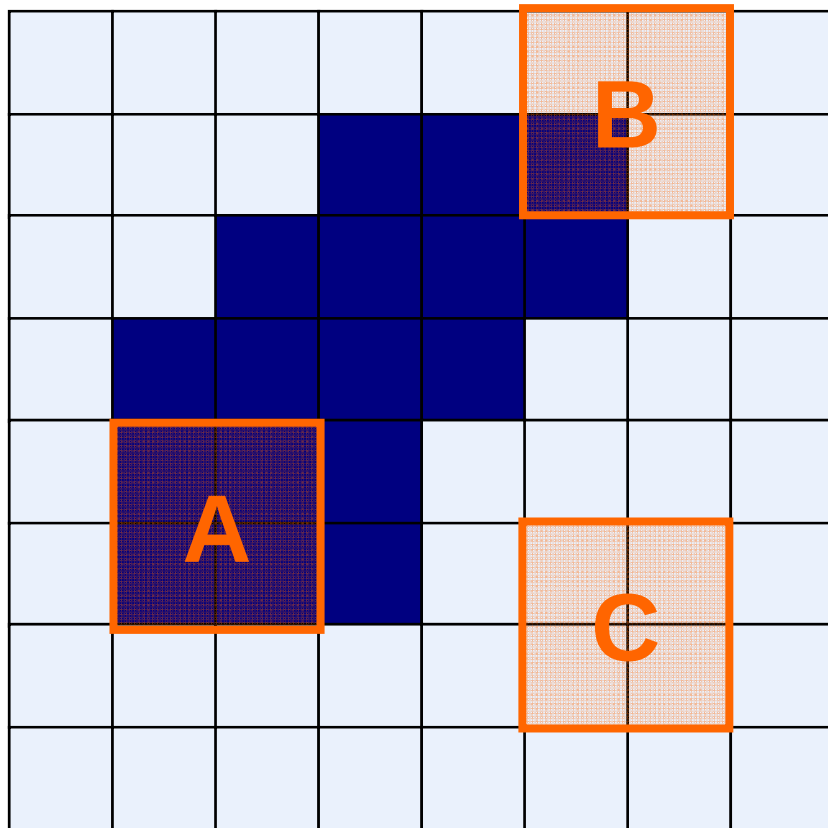


FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Structuring Elements, Hits & Fits



Structuring Element

Fit: All *on pixels* in the structuring element cover *on pixels* in the image

Hit: Any *on pixel* in the structuring element covers an *on pixel* in the image

All morphological processing operations are based on these simple ideas

Structuring Elements

Structuring elements can be any size and make any shape

However, for simplicity we will use rectangular structuring elements with their origin at the middle pixel

1	1	1
1	1	1
1	1	1

0	1	0
1	1	1
0	1	0

0	0	1	0	0
0	1	1	1	0
1	1	1	1	1
0	1	1	1	0
0	0	1	0	0

Fitting & Hitting

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0	0	0
0	0	1	B	1	1	1	0	0	C	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	A	1	1	1	0
0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0

1	1	1
1	1	1
1	1	1

Structuring
Element 1

0	1	0
1	1	1
0	1	0

Structuring
Element 2

Basic morphological operations

- Erosion

shrink

- Dilation

grow

- combine to

- Opening → object
- Closing → background

keep general shape but
smooth with respect to

Erosion

- Does the structuring element **fit the set?**

erosion of a set A by structuring element B :
all z in A such that B is in A when origin of
 $B=z$

$$A \ominus B = \{z \mid (B)_z \subseteq A\}$$

shrink the object

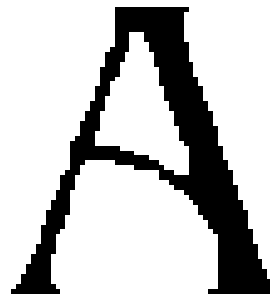
Erosion

Erosion of image A by structuring element B is given by $A \ominus B$. The structuring element B is positioned with its origin at (x, y) and the new pixel value is determined using the rule:

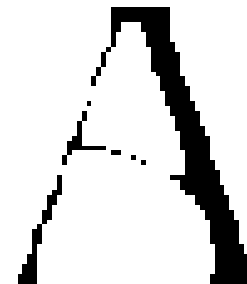
$$g(x, y) = \begin{cases} 1 & \text{if } B \text{ fits } A \\ 0 & \text{otherwise} \end{cases}$$



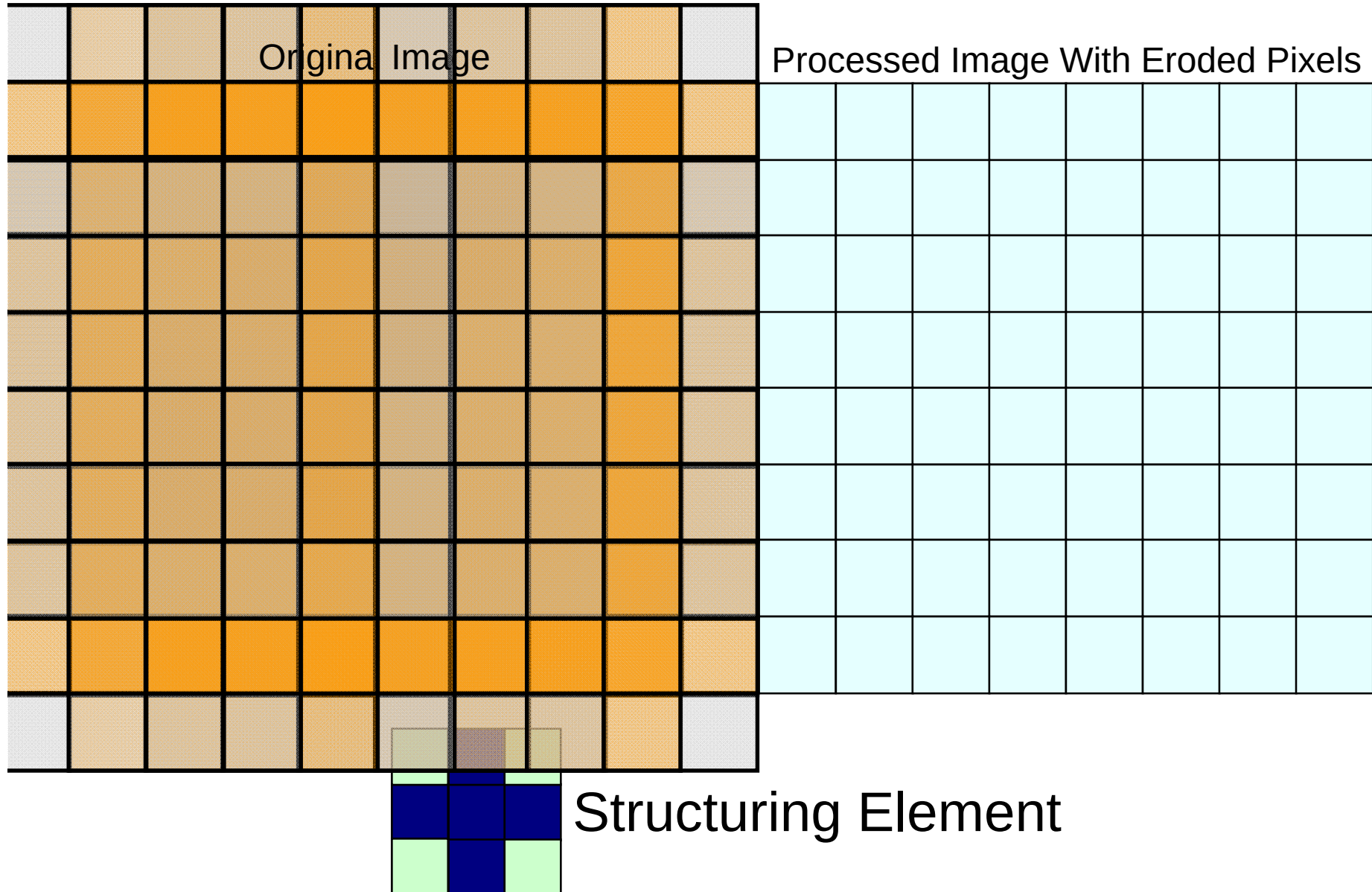
Original image



Erosion by 3*3
square structuring
element

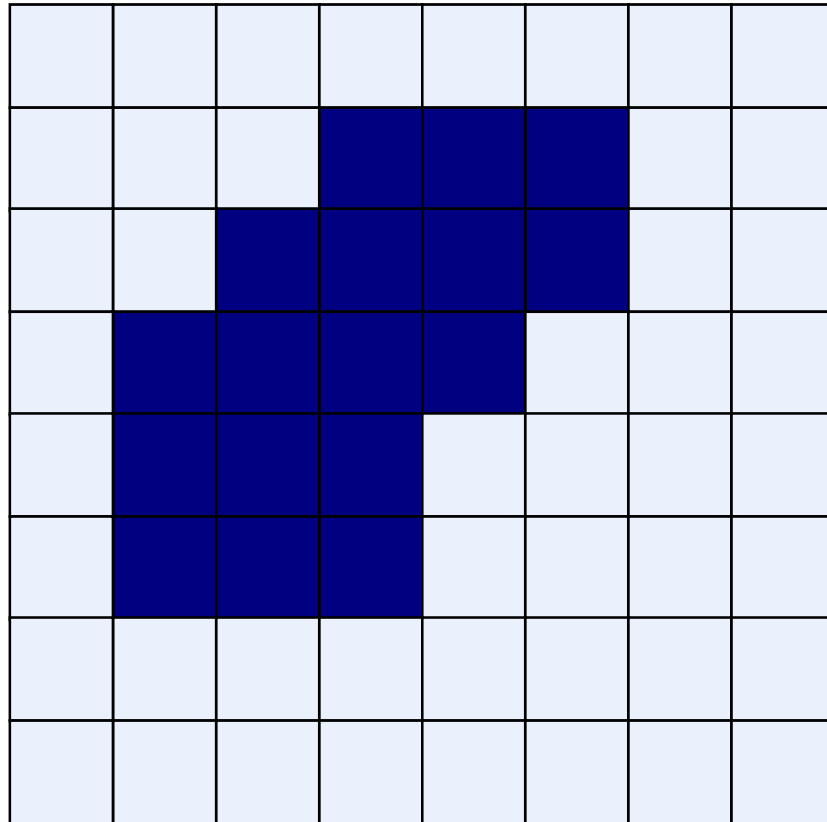


Erosion by 5*5 square
structuring element

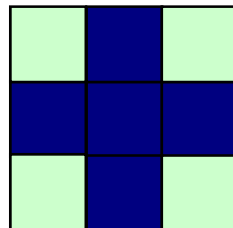
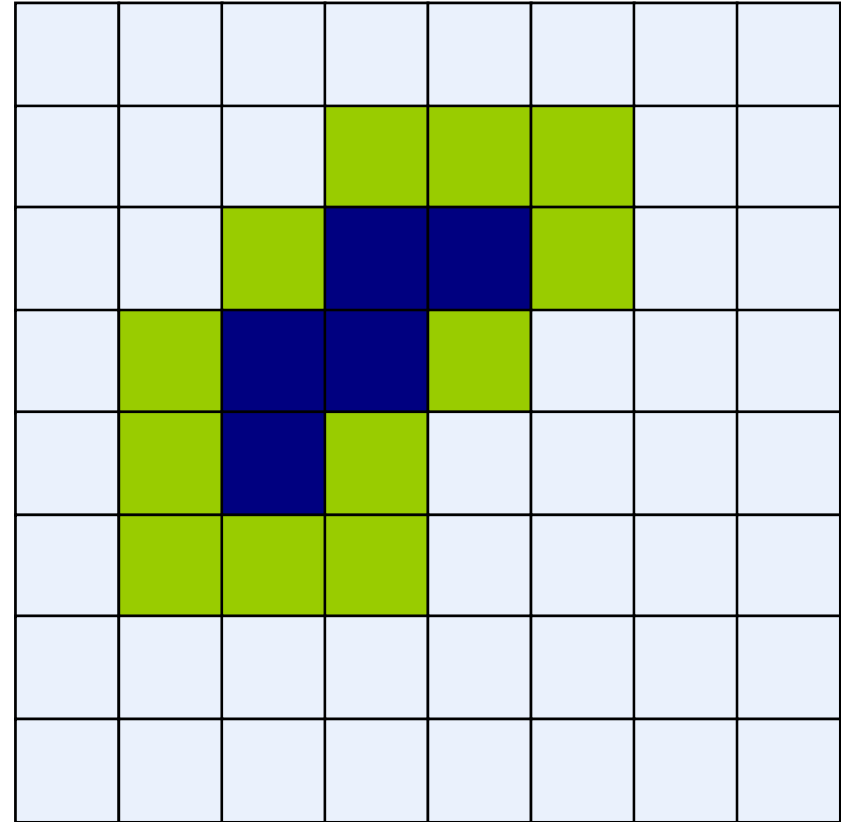


Erosion Example

Original Image



Processed Image



Structuring Element

Erosion

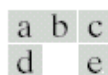
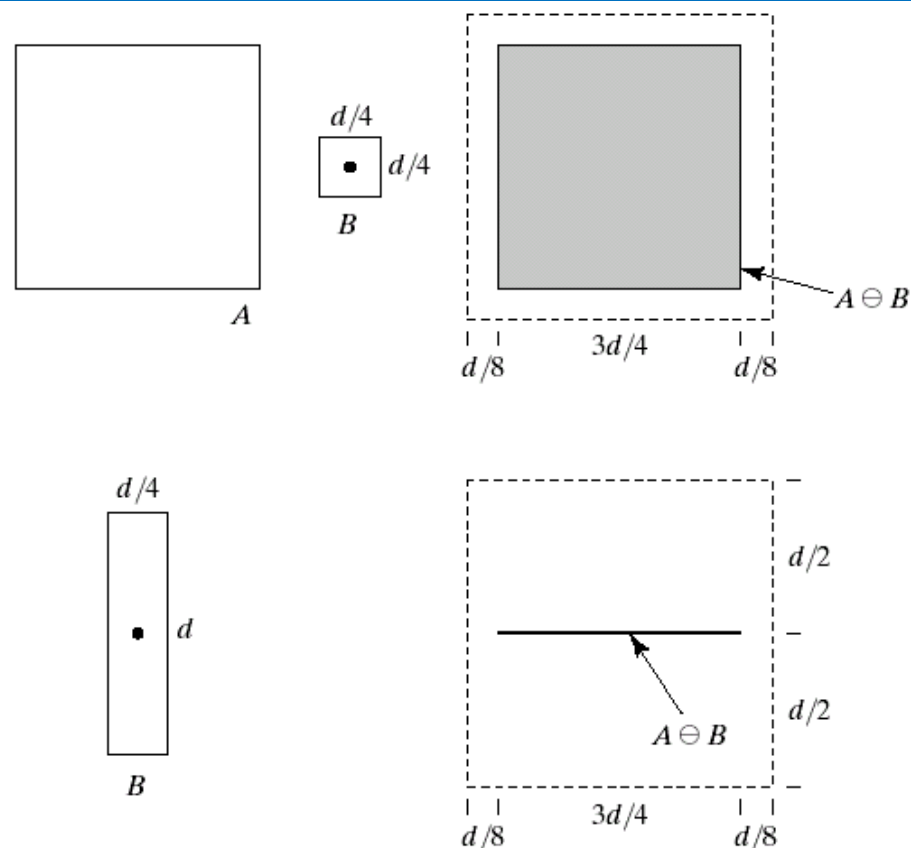
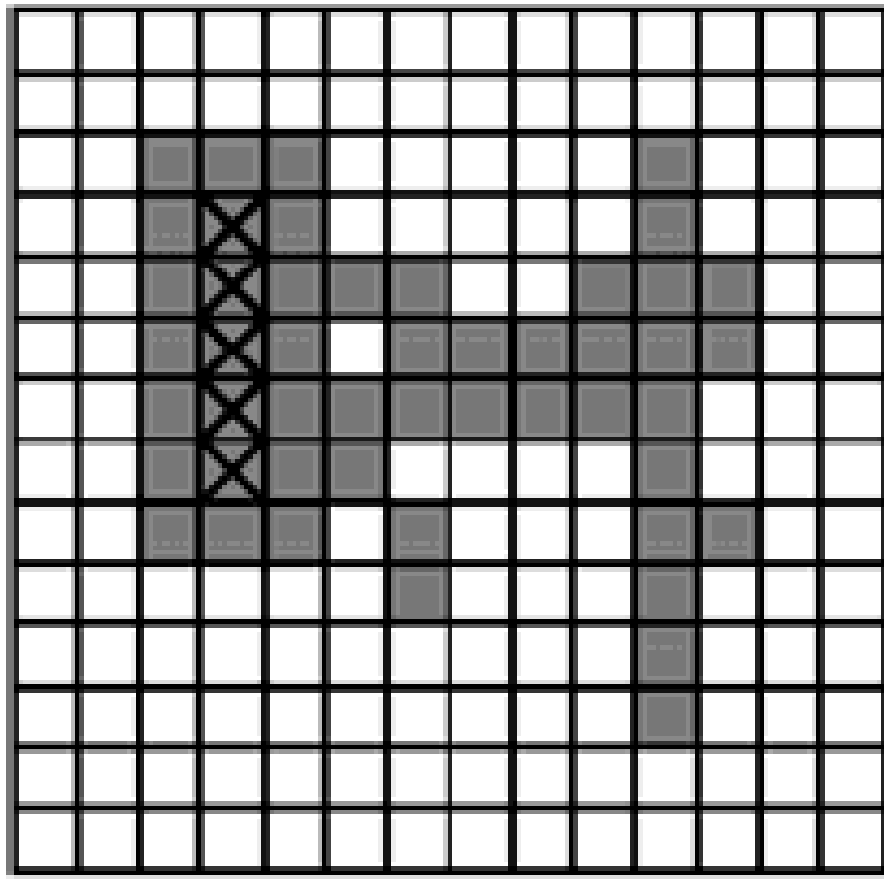


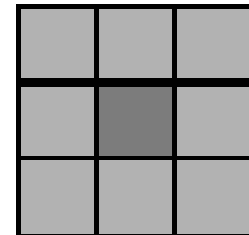
FIGURE 9.6 (a) Set A. (b) Square structuring element. (c) Erosion of A by B, shown shaded. (d) Elongated structuring element. (e) Erosion of A using this element.

$$A \ominus B = \{z | (B)_z \subseteq A\}$$

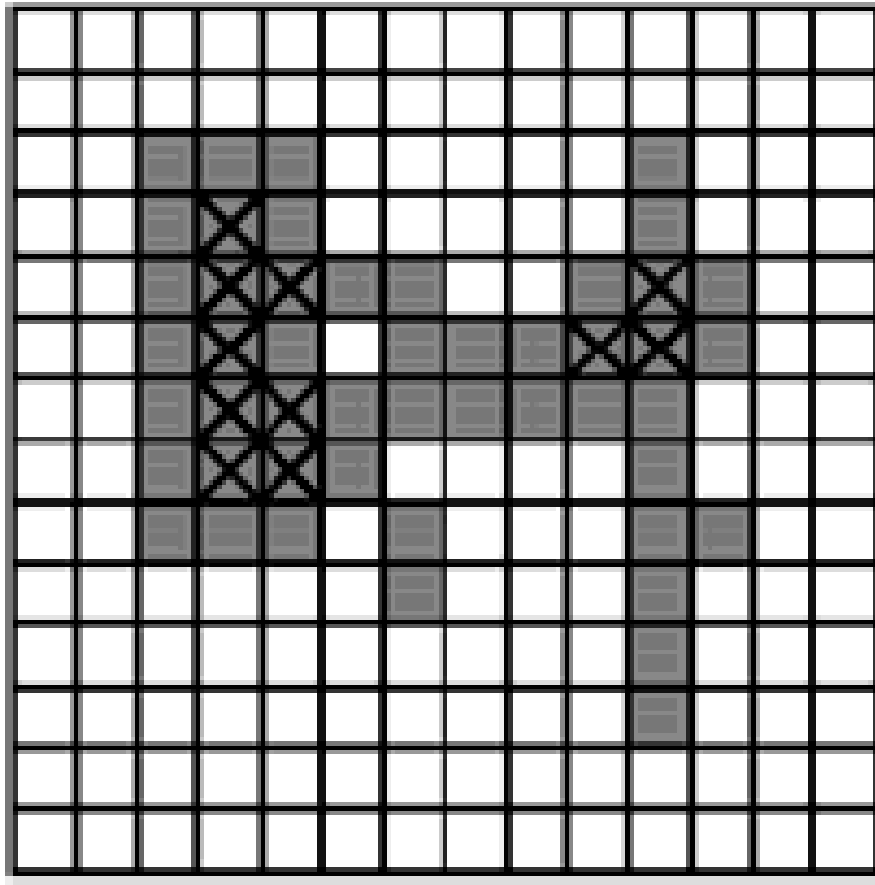
Erosion



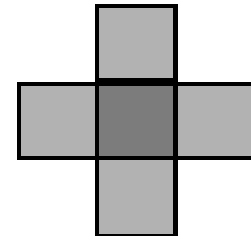
SE=



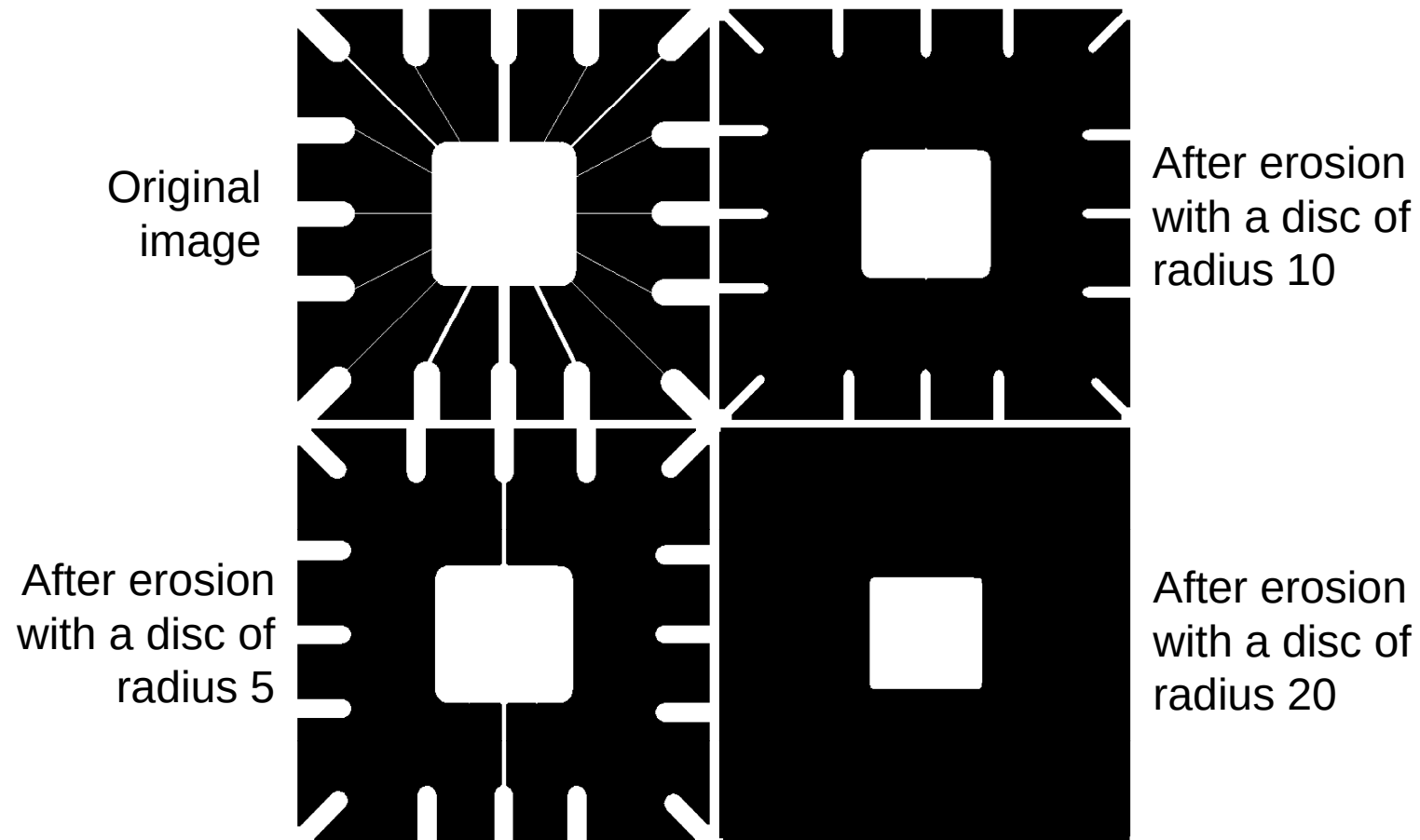
Erosion



SE=



Erosion Example



Erosion : eliminating irrelevant detail



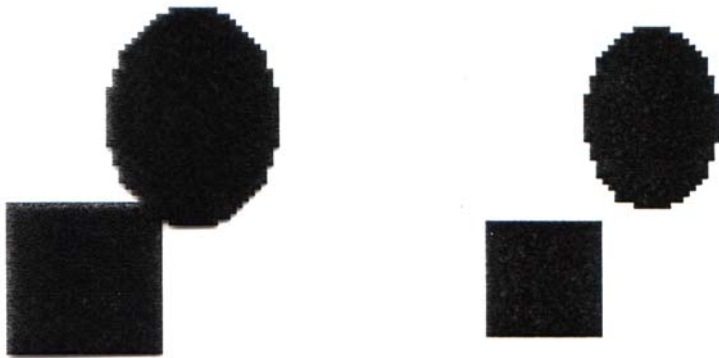
a b c

FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.

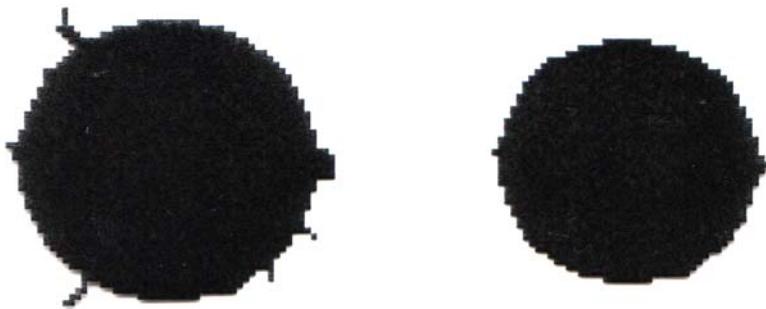
structuring element $B = 13 \times 13$ pixels of gray level 1

What Is Erosion For?

Erosion can split apart joined objects



Erosion can strip away extrusions



Watch out: Erosion shrinks objects

Dilation

- Does the structuring element **hit the set**?
- dilation of a set A by structuring element B :
all z in A such that B hits A when origin of $B=z$

$$A \oplus B = \{z | (\hat{B})_z \cap A \neq \Phi\}$$

- **grow the object**

Dilation

Dilation of image A by structuring element B is given by $A \oplus B$. The structuring element s is positioned with its origin at (x, y) and the new pixel value is determined using the rule:

$$g(x, y) = \begin{cases} 1 & \text{if } B \text{ hits } A \\ 0 & \text{otherwise} \end{cases}$$



Original image

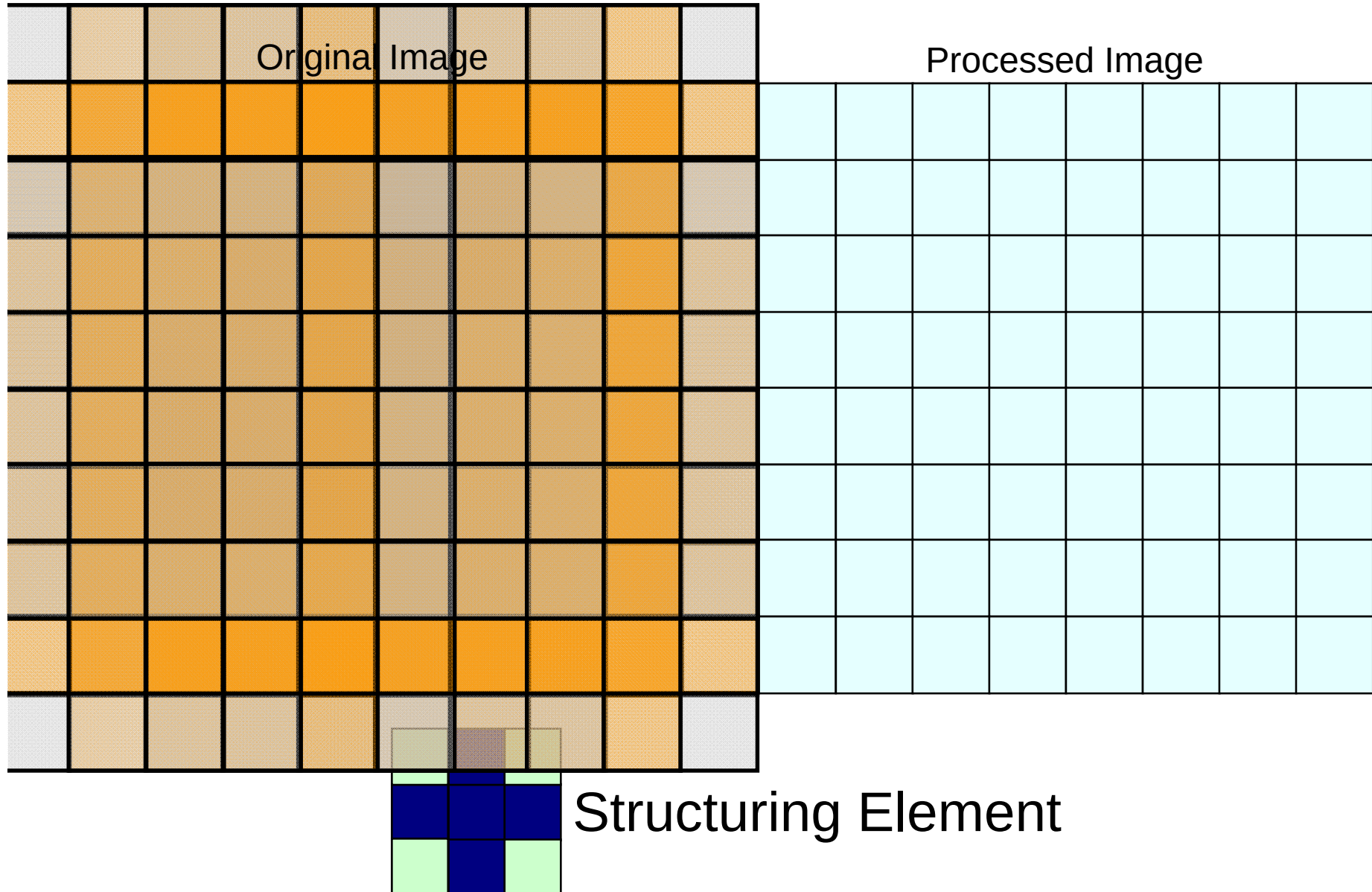


Dilation by 3*3
square structuring
element



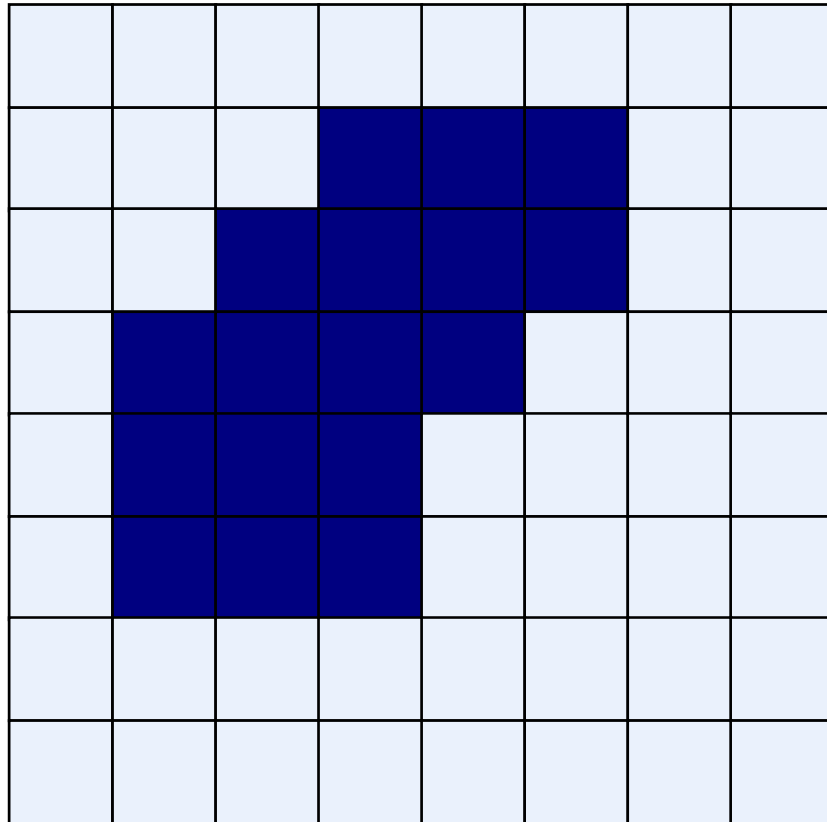
Dilation by 5*5 square
structuring element

Dilation Example

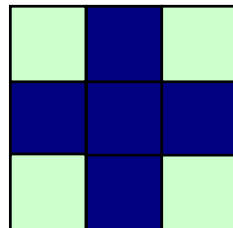
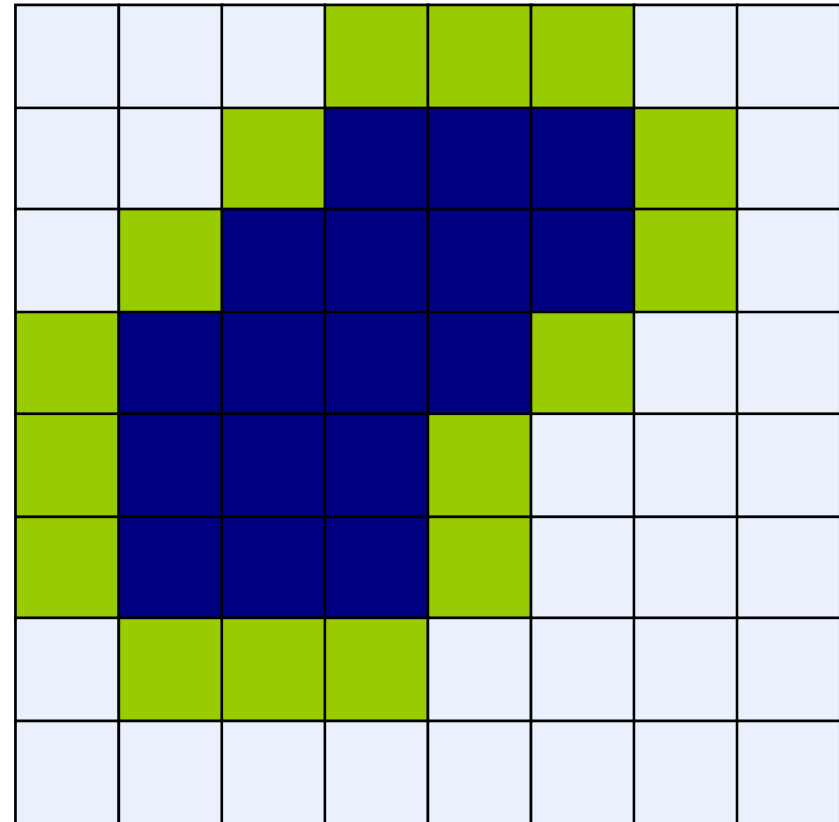


Dilation Example

Original Image



Processed Image With Dilated Pixels



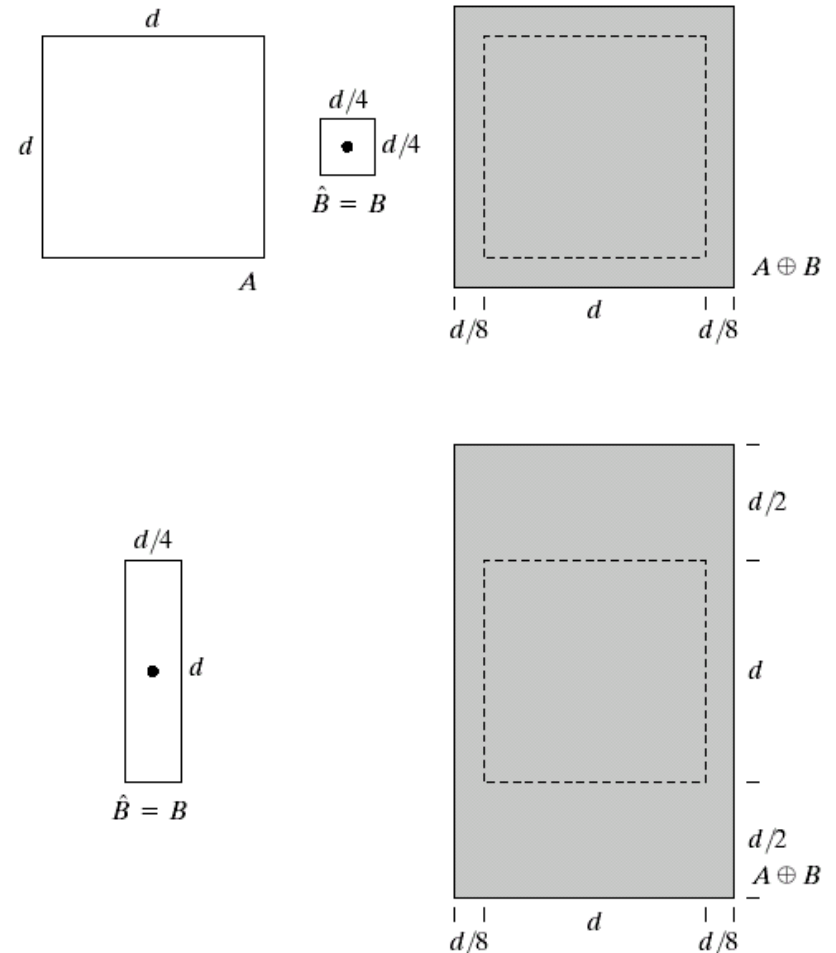
Structuring Element

Dilation

a	b	c
d	e	

FIGURE 9.4

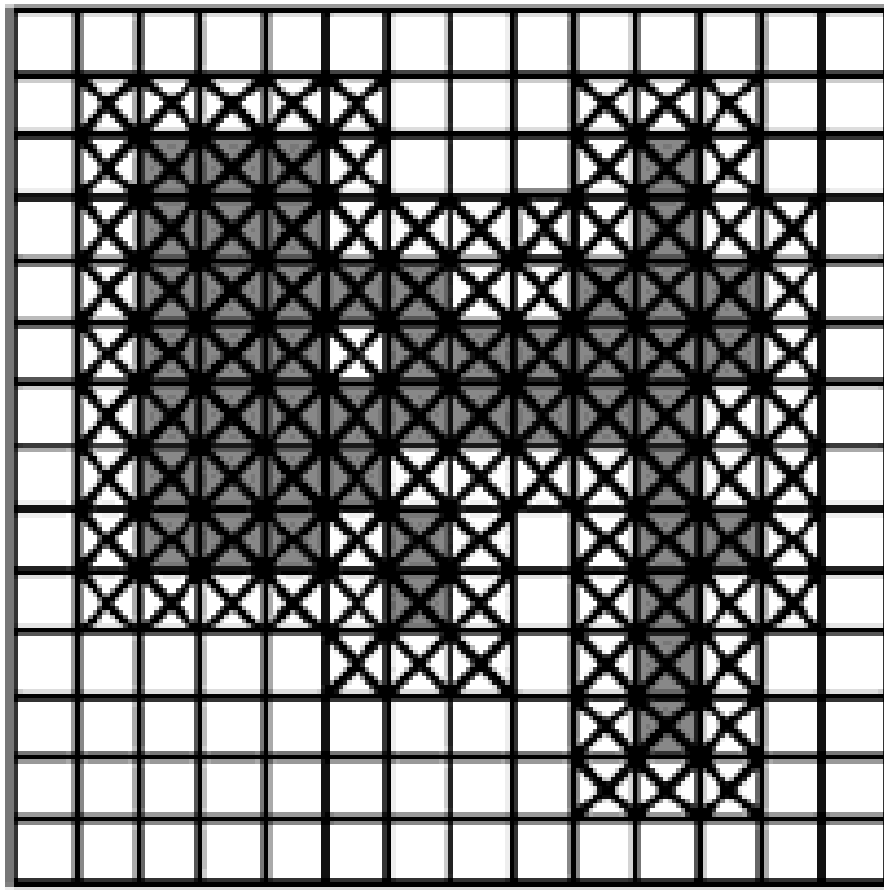
- (a) Set A .
 (b) Square structuring element (dot is the center).
 (c) Dilation of A by B , shown shaded.
 (d) Elongated structuring element.
 (e) Dilation of A using this element.



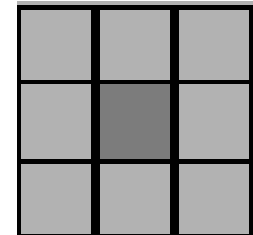
B = structuring element

$$A \oplus B = \{z | (\hat{B})_z \cap A \neq \Phi\}$$

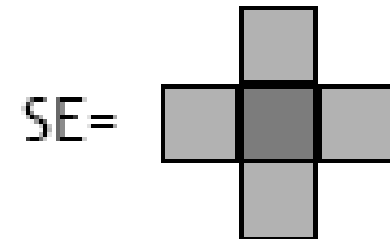
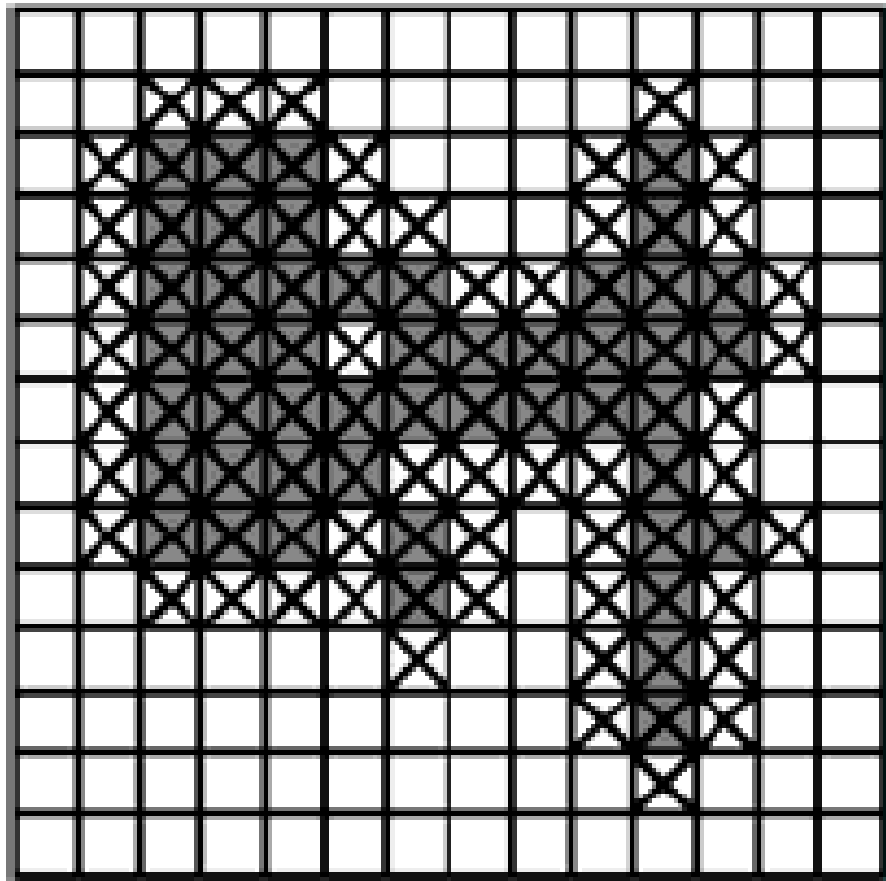
Dilation



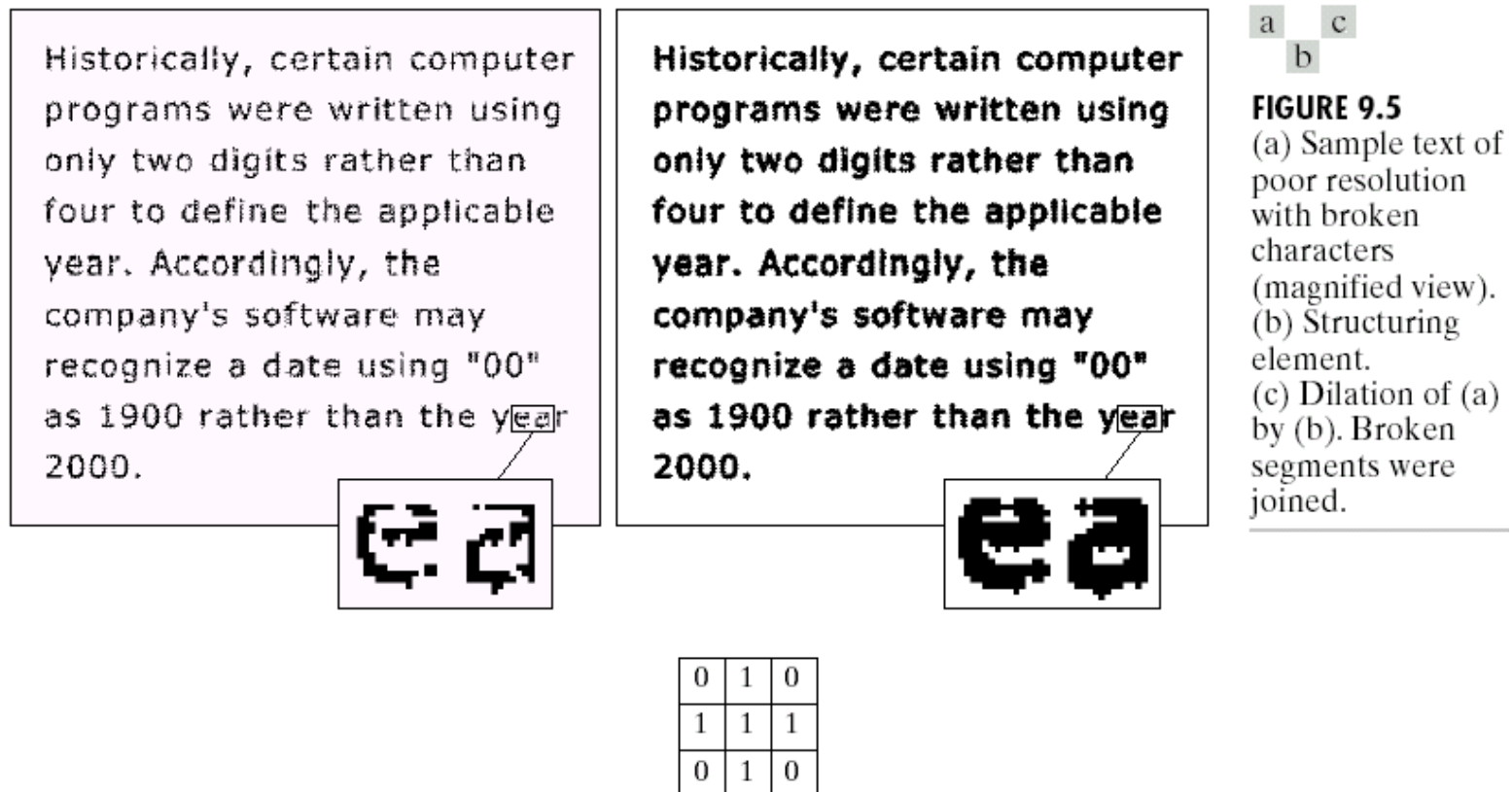
SE=



Dilation



Dilation : Bridging gaps



What Is Dilation For?

Dilation can repair breaks



Dilation can repair intrusions



Watch out: Dilation enlarges objects

useful

- erosion
 - removal of structures of certain shape and size, given by SE
- Dilation
 - filling of holes of certain shape and size, given by SE

Combining erosion and dilation

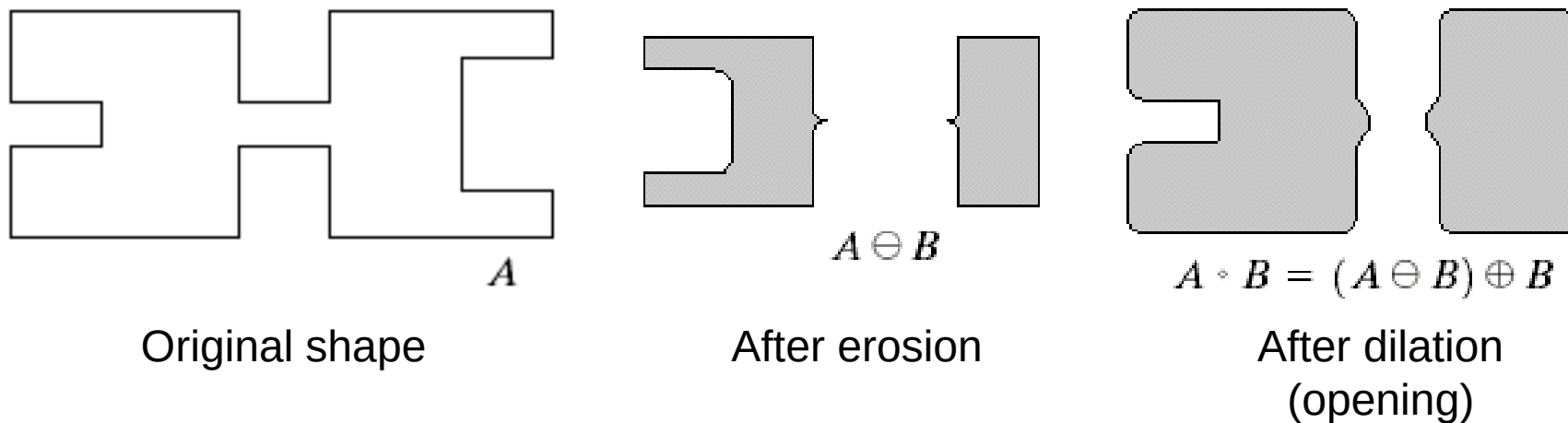
- WANTED:
 - remove structures / fill holes
 - without affecting remaining parts
- SOLUTION:
- combine erosion and dilation
- (using same SE)

Opening

erosion followed by dilation, denoted $\circ$

$$A \circ B = (A \ominus B) \oplus B$$

- eliminates protrusions
- breaks necks
- smoothes contour



Opening Example

Original
Image

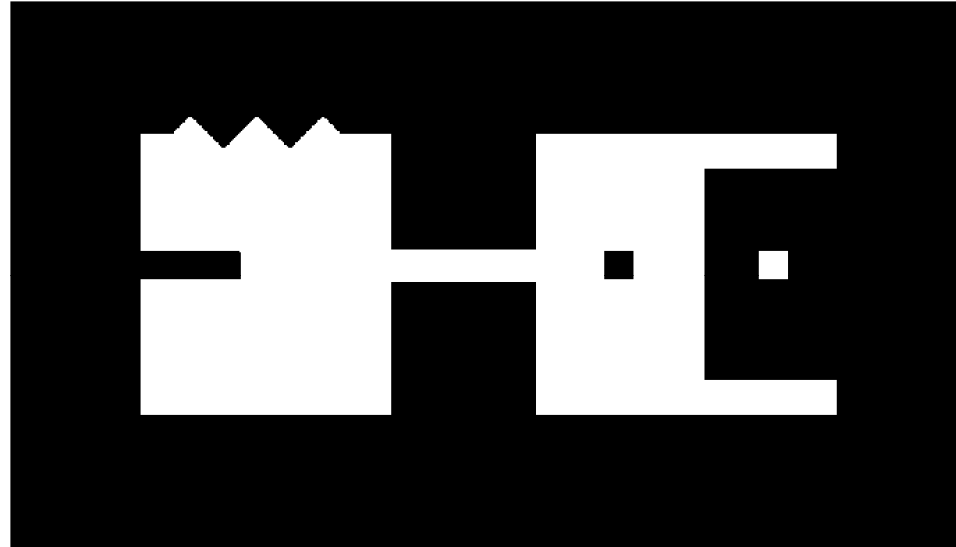
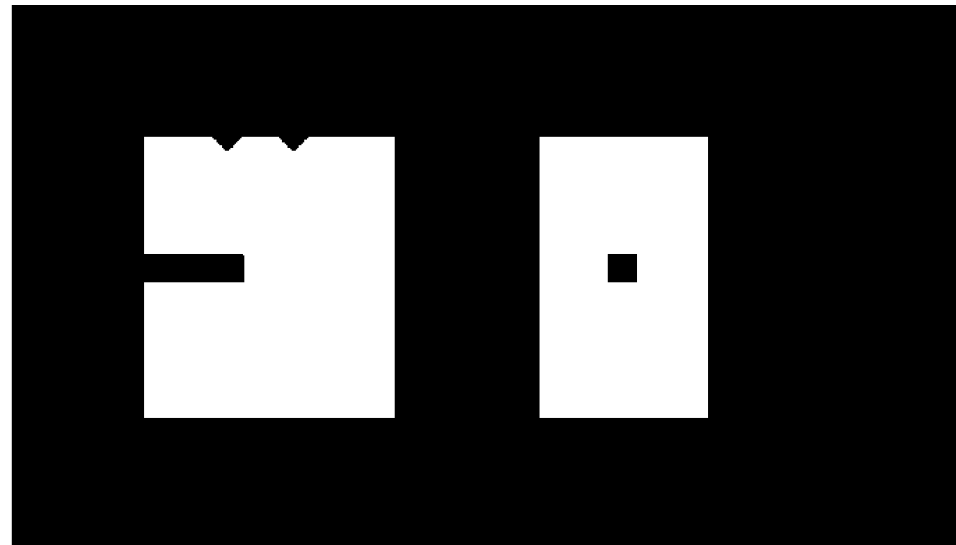
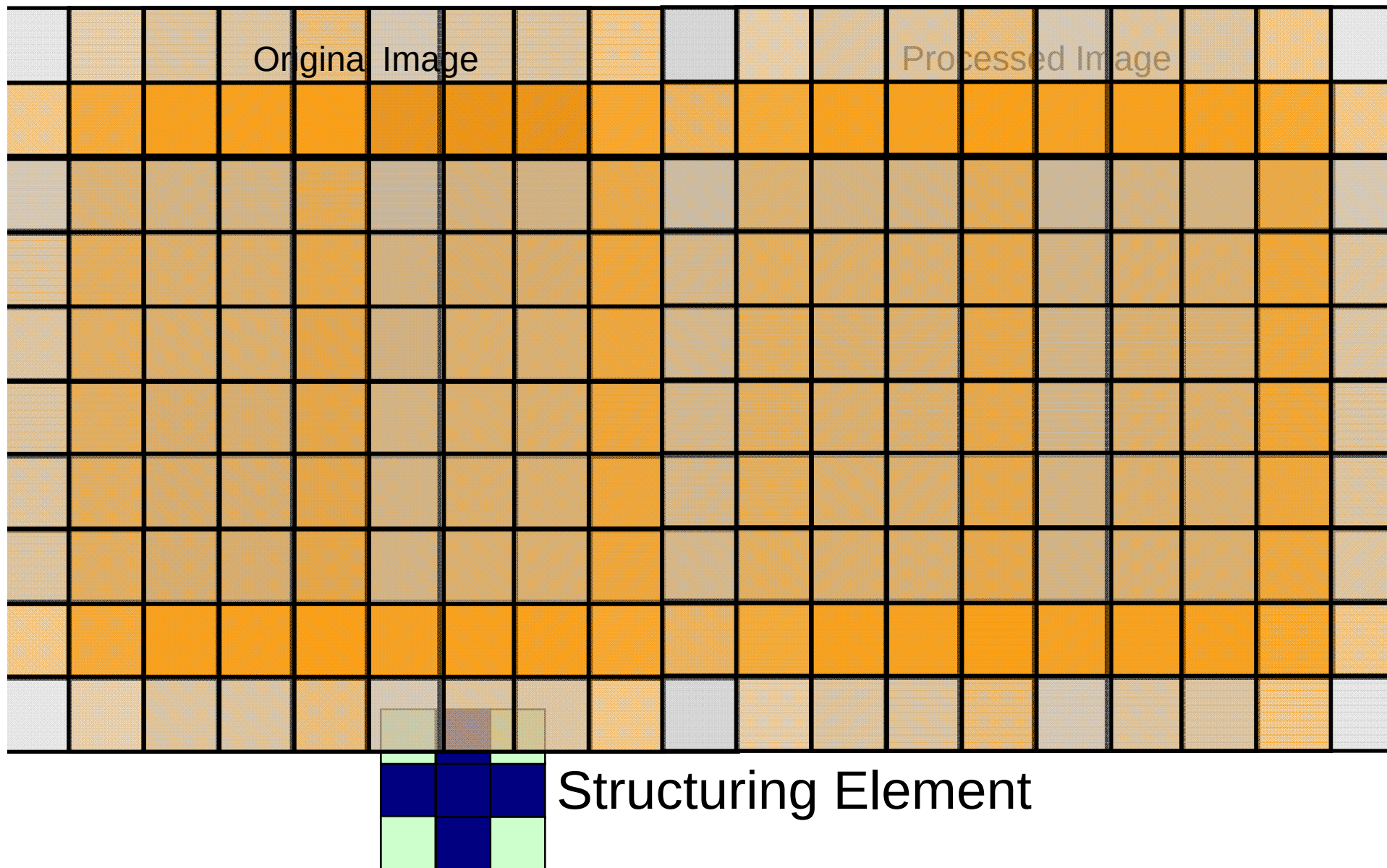


Image
After
Opening

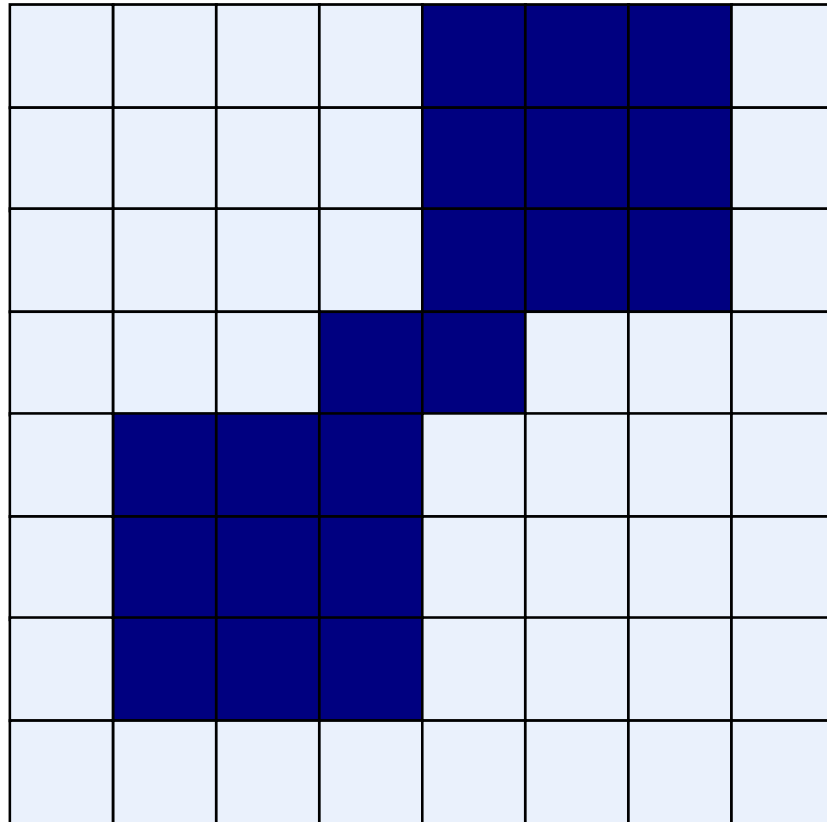


Opening Example

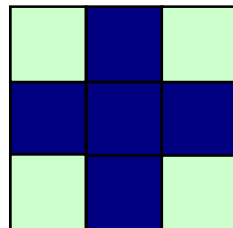
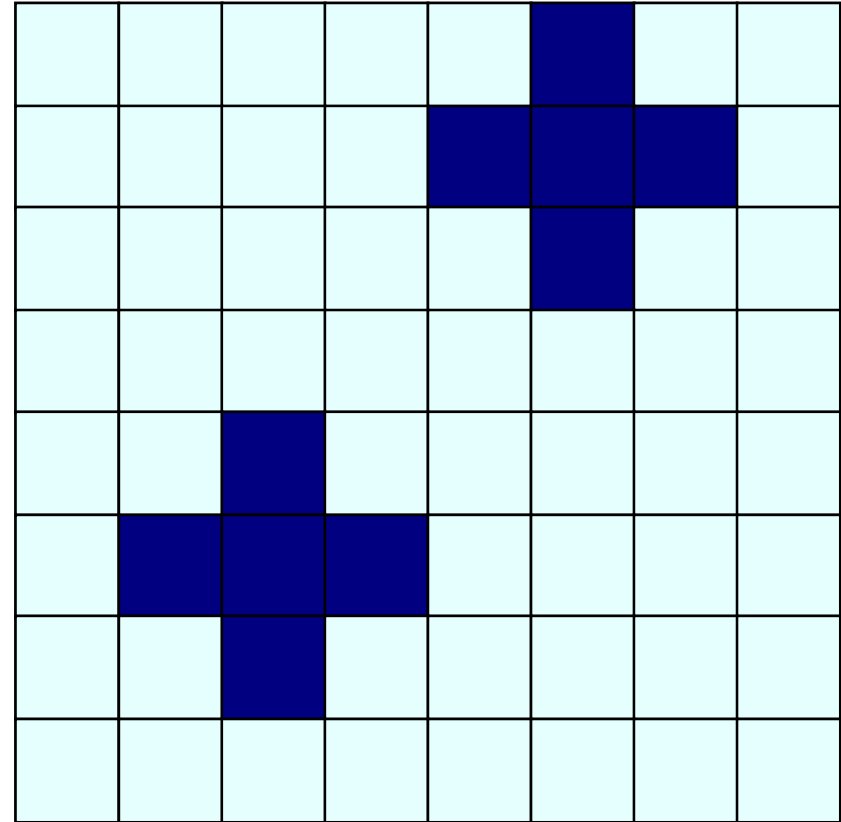


Opening Example

Original Image

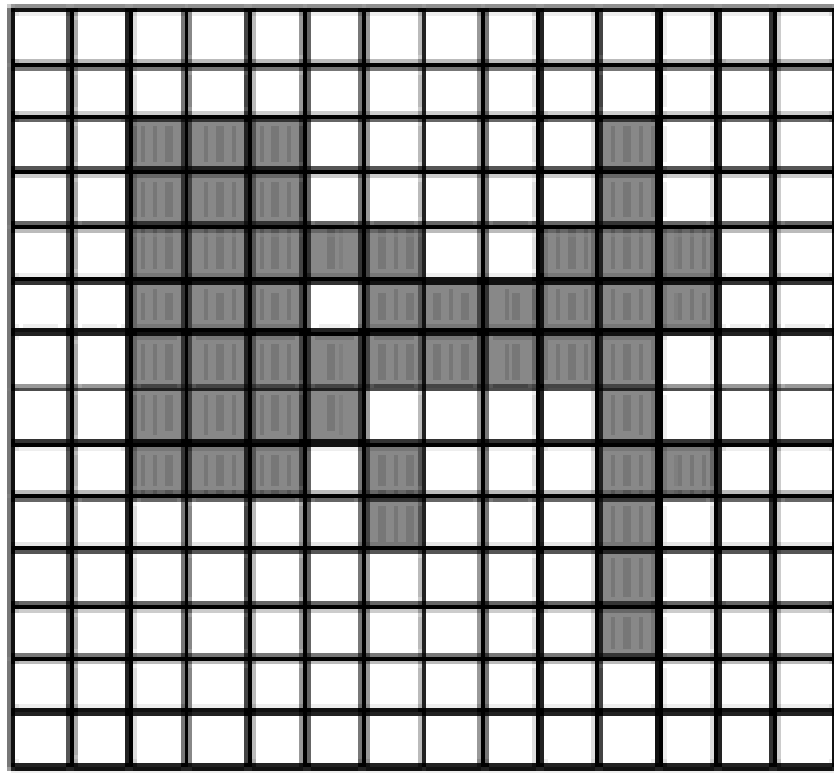
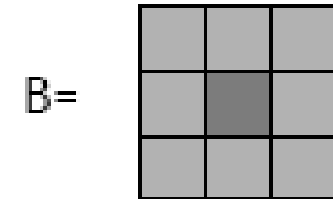


Processed Image

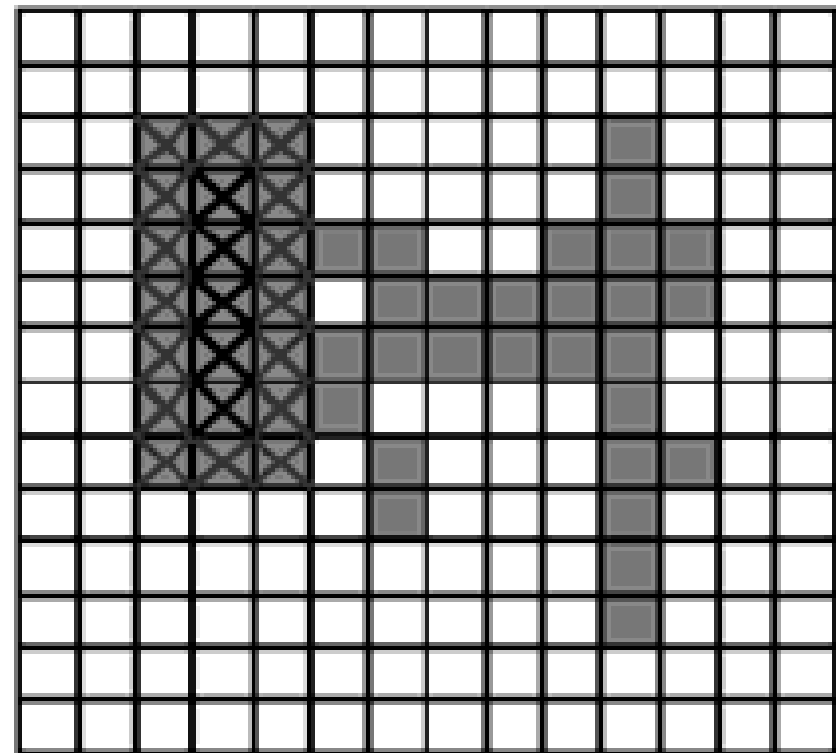


Structuring Element

Opening

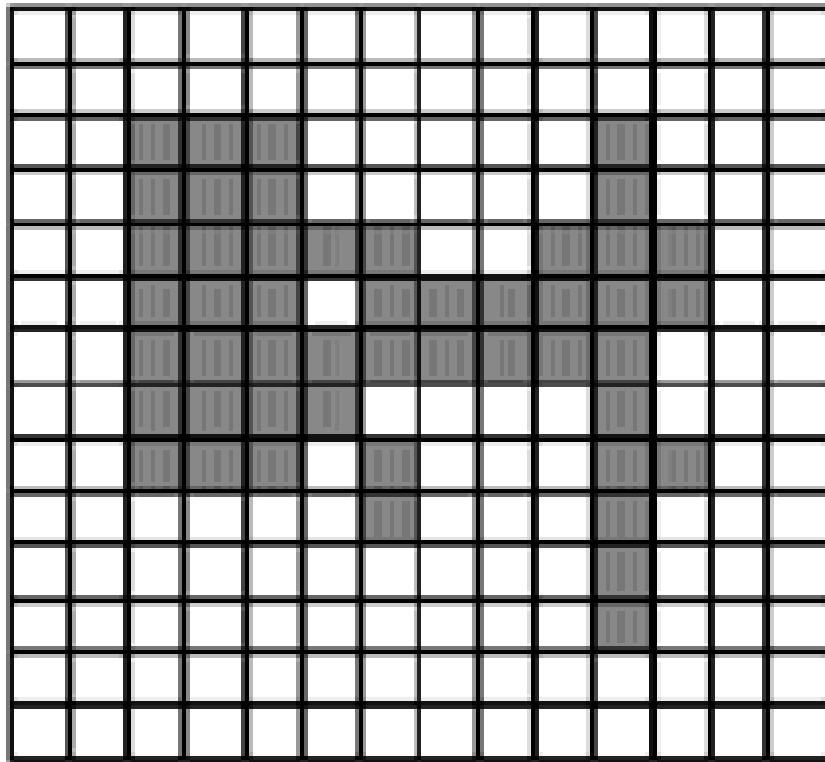
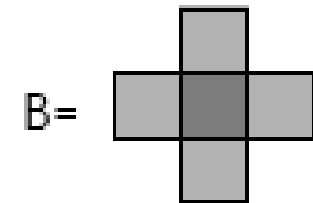


A

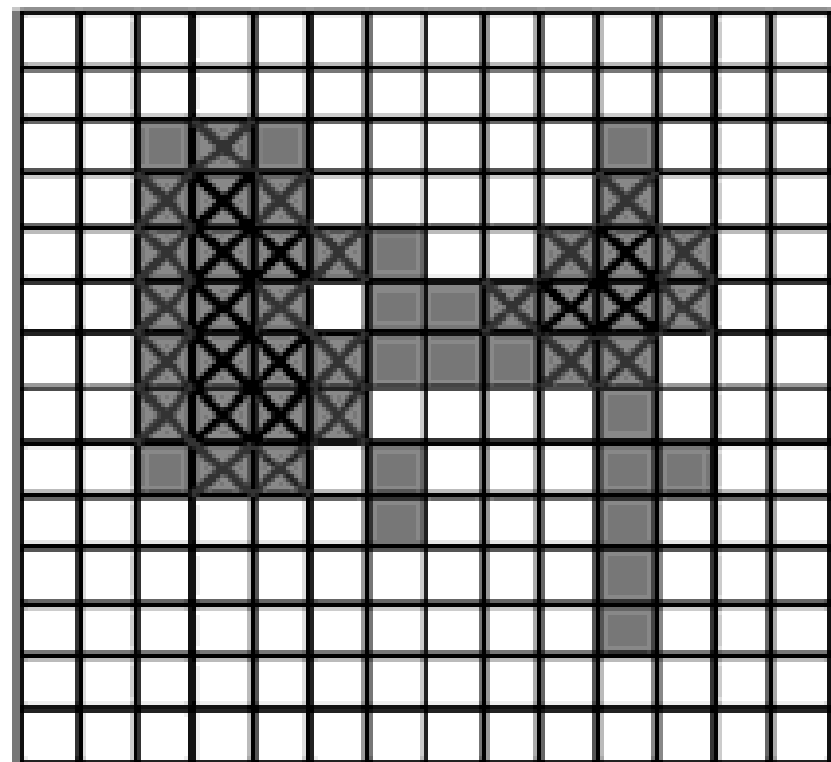


$A \ominus B$ $A \circ B$

Opening



A



$A \ominus B$ $A \circ B$

Opening

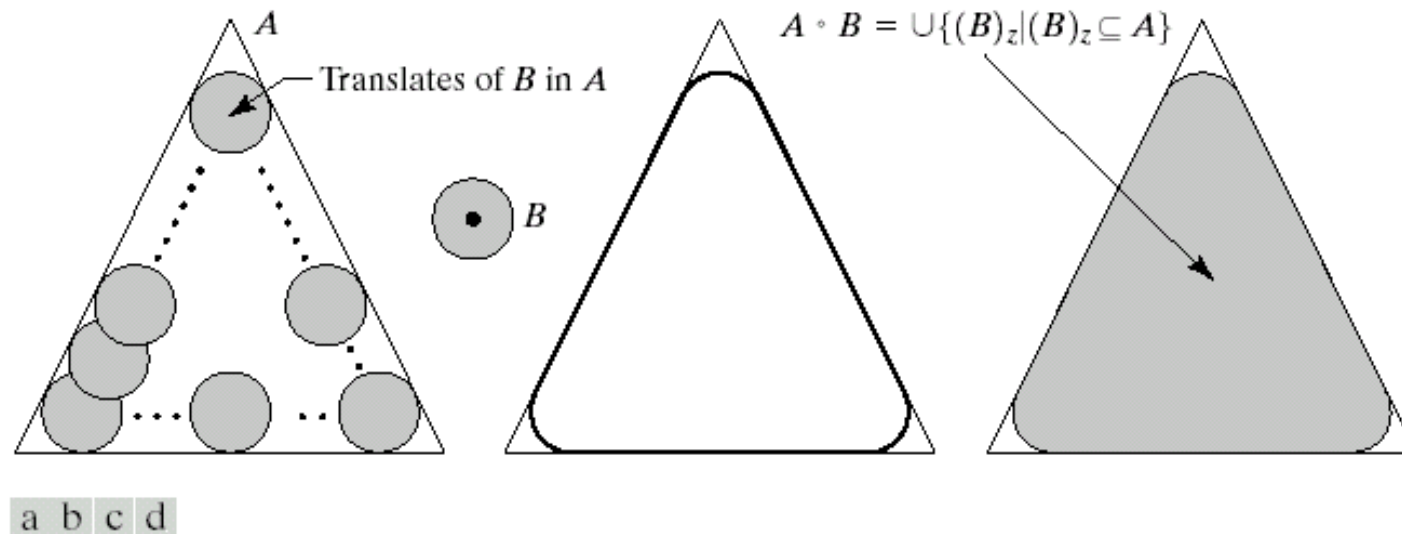


FIGURE 9.8 (a) Structuring element B “rolling” along the inner boundary of A (the dot indicates the origin of B). (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded).

$$A \circ B = (A \ominus B) \oplus B$$

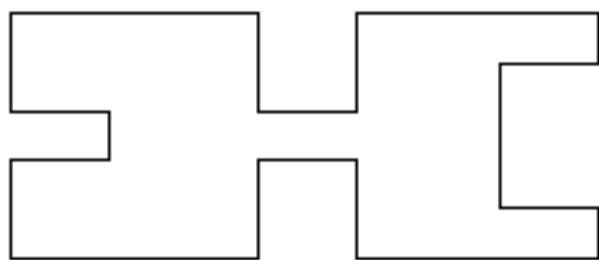
$$A \circ B = \cup \{(B)_z \mid (B)_z \subseteq A\}$$

Closing

dilation followed by erosion, denoted •

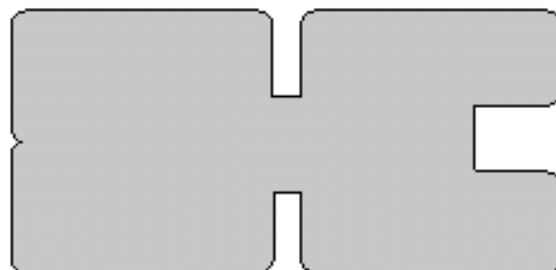
$$A \bullet B = (A \oplus B) \ominus B$$

- smooth contour
- fuse narrow breaks and long thin gulfs
- eliminate small holes
- fill gaps in the contour



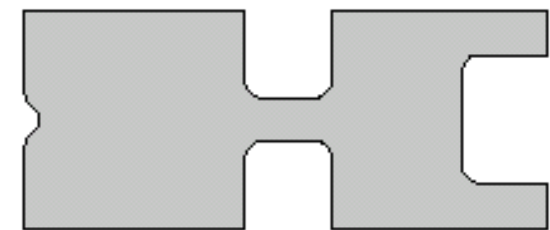
A

Original shape



$A \oplus B$

After dilation



$A \bullet B = (A \oplus B) \ominus B$

After erosion
(closing)

Closing Example

Original
Image

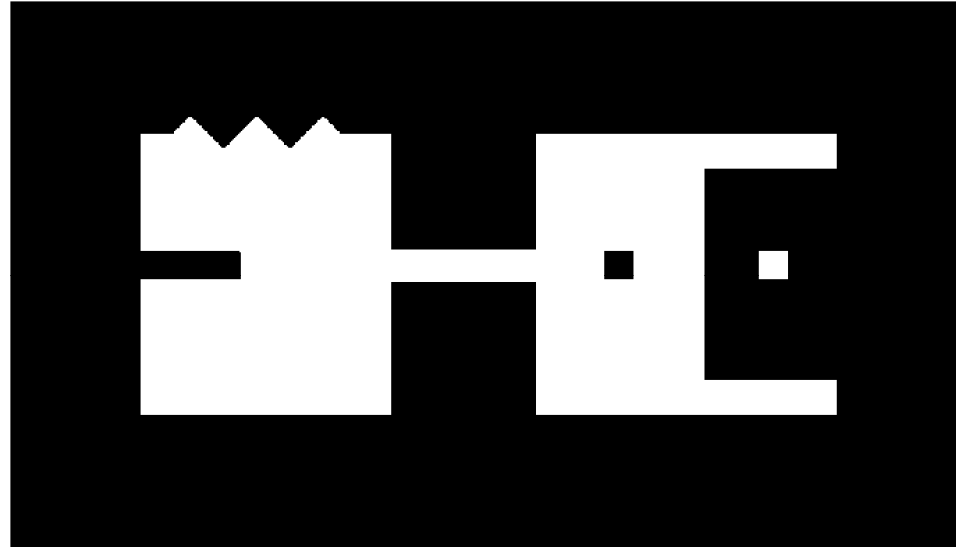
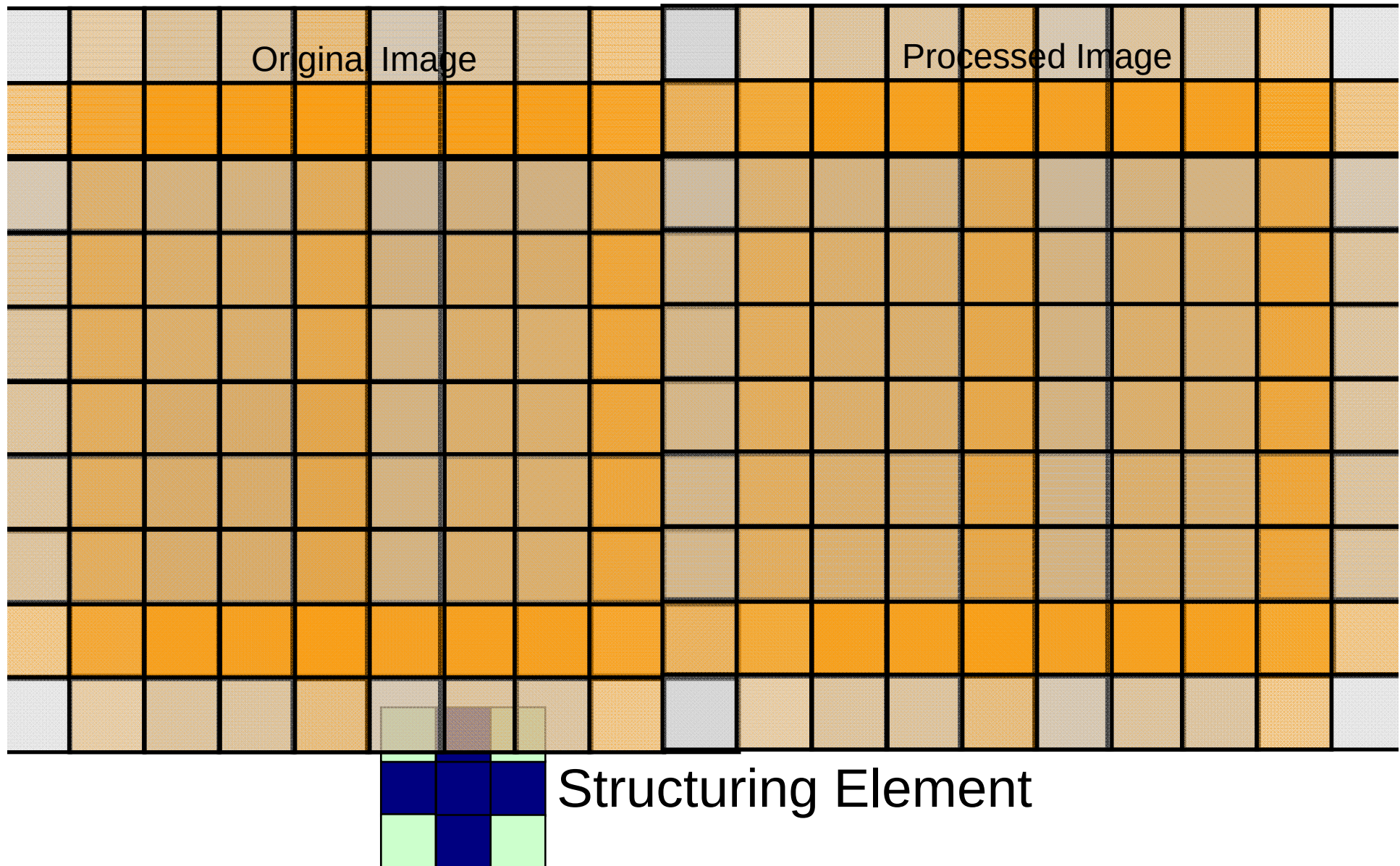


Image
After
Closing

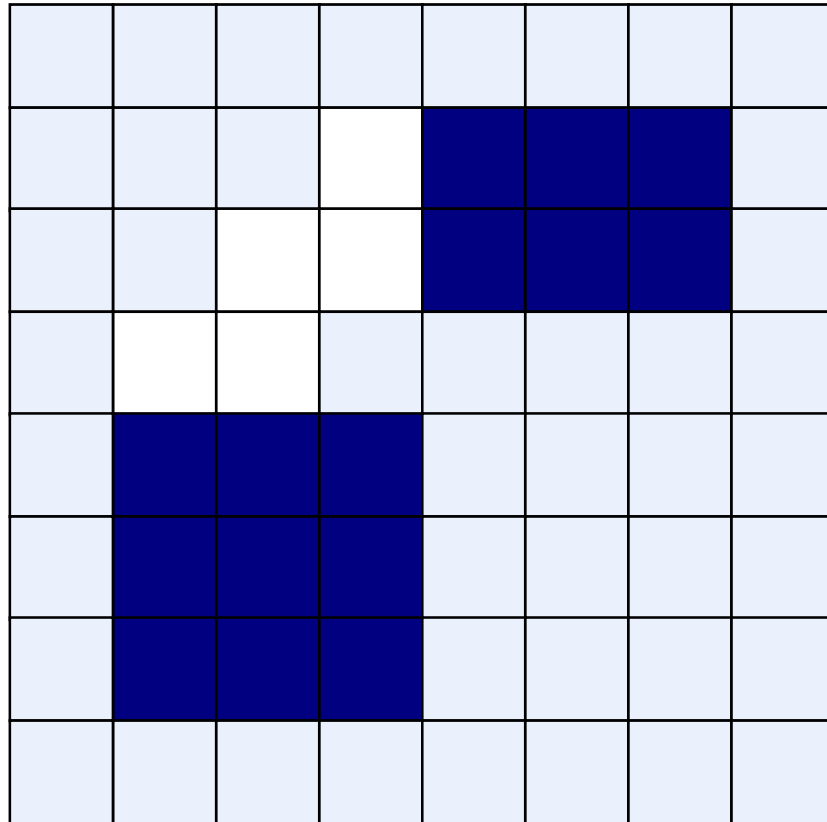


Closing Example

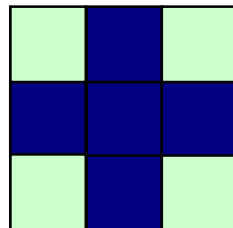
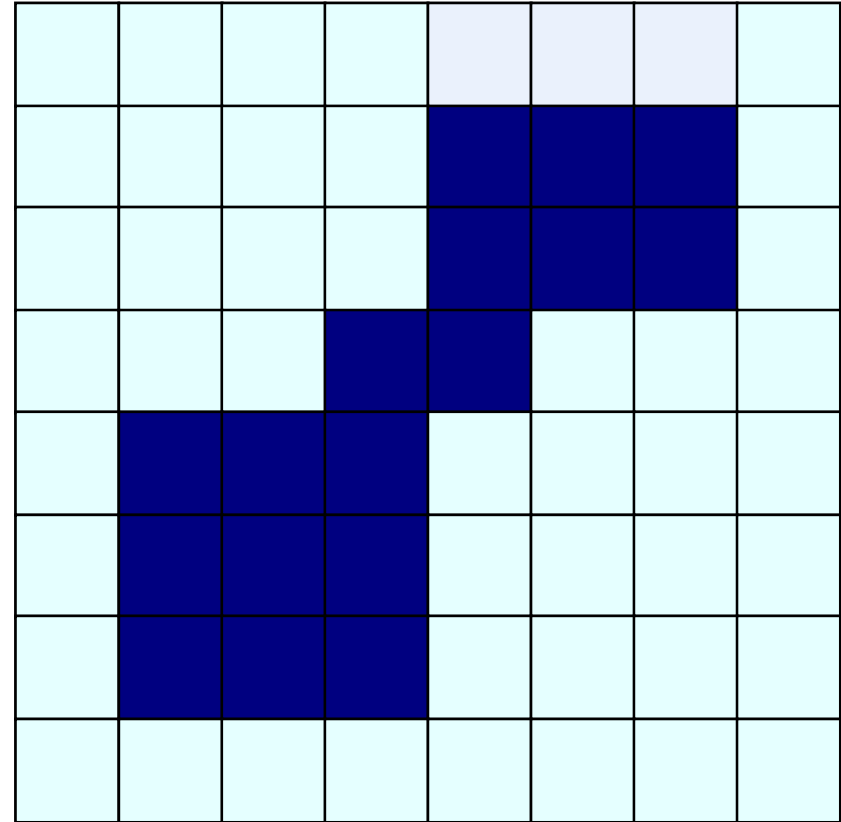


Closing Example

Original Image



Processed Image

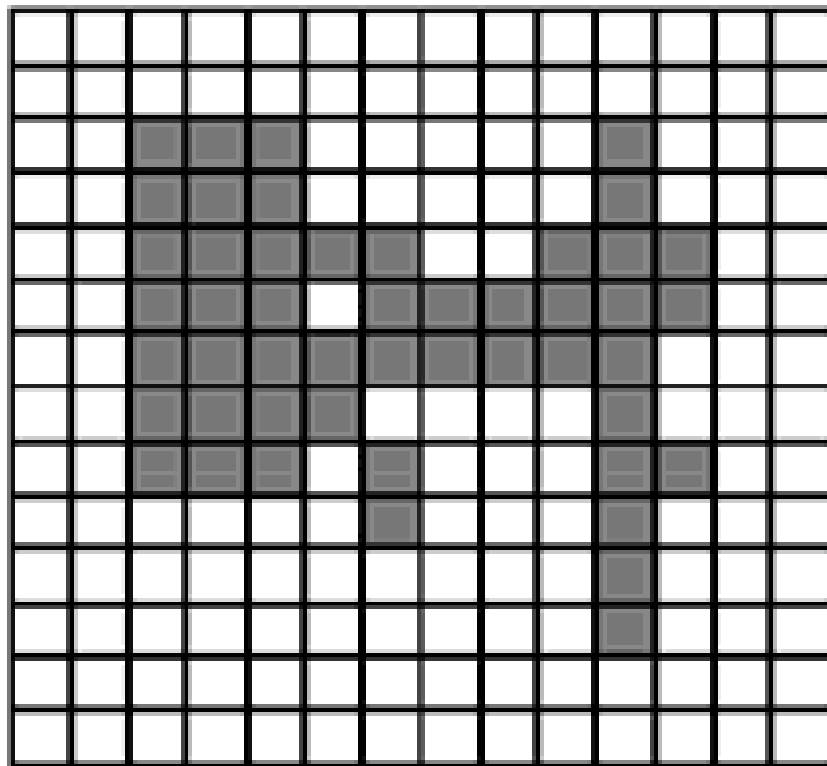
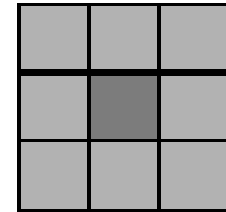


Structuring Element

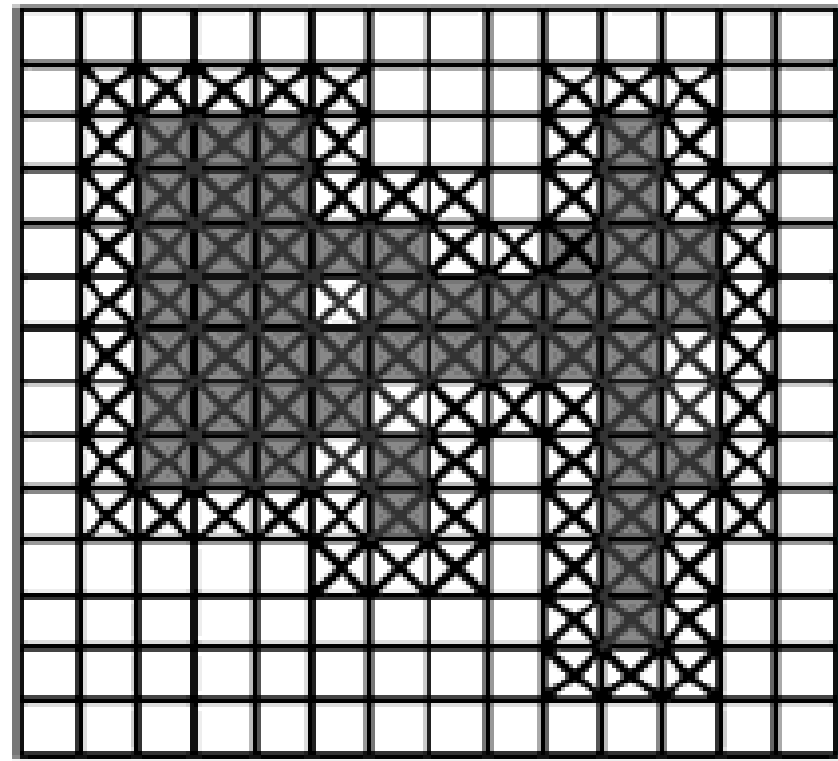
Closing



$B =$

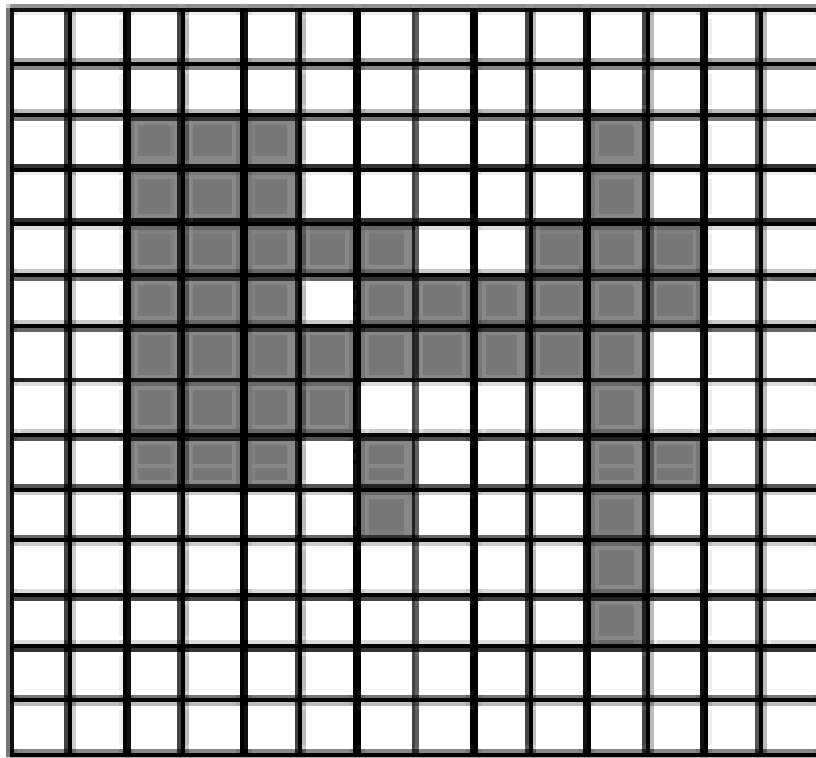
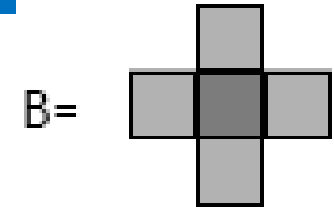


A

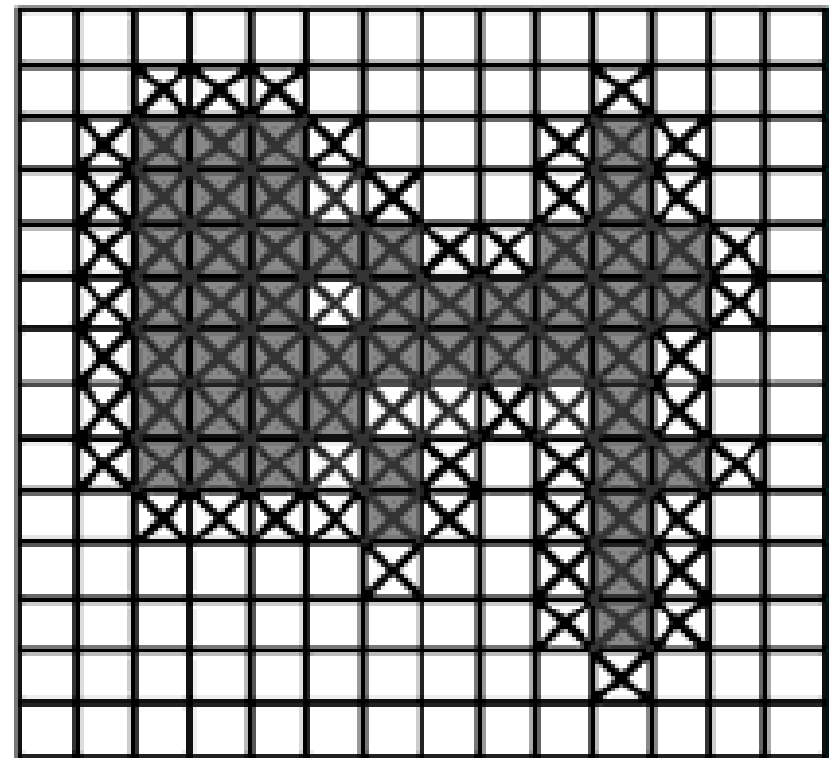


$A \oplus B \quad A \cdot B$

Closing



A



$A \oplus B \quad A \cdot B$

Closing

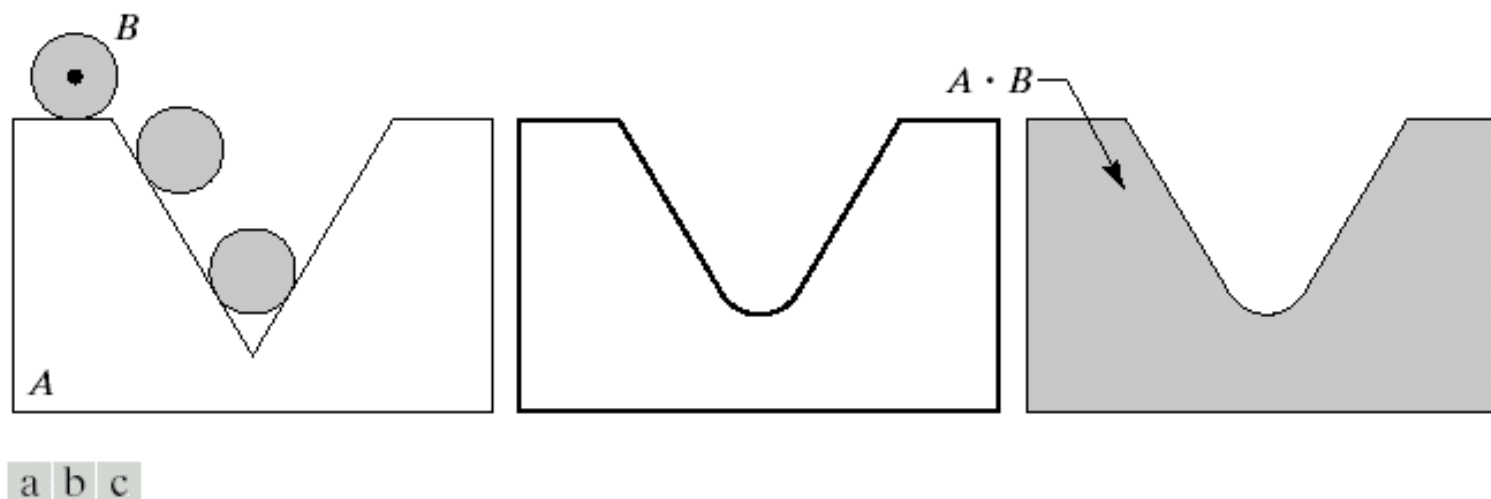
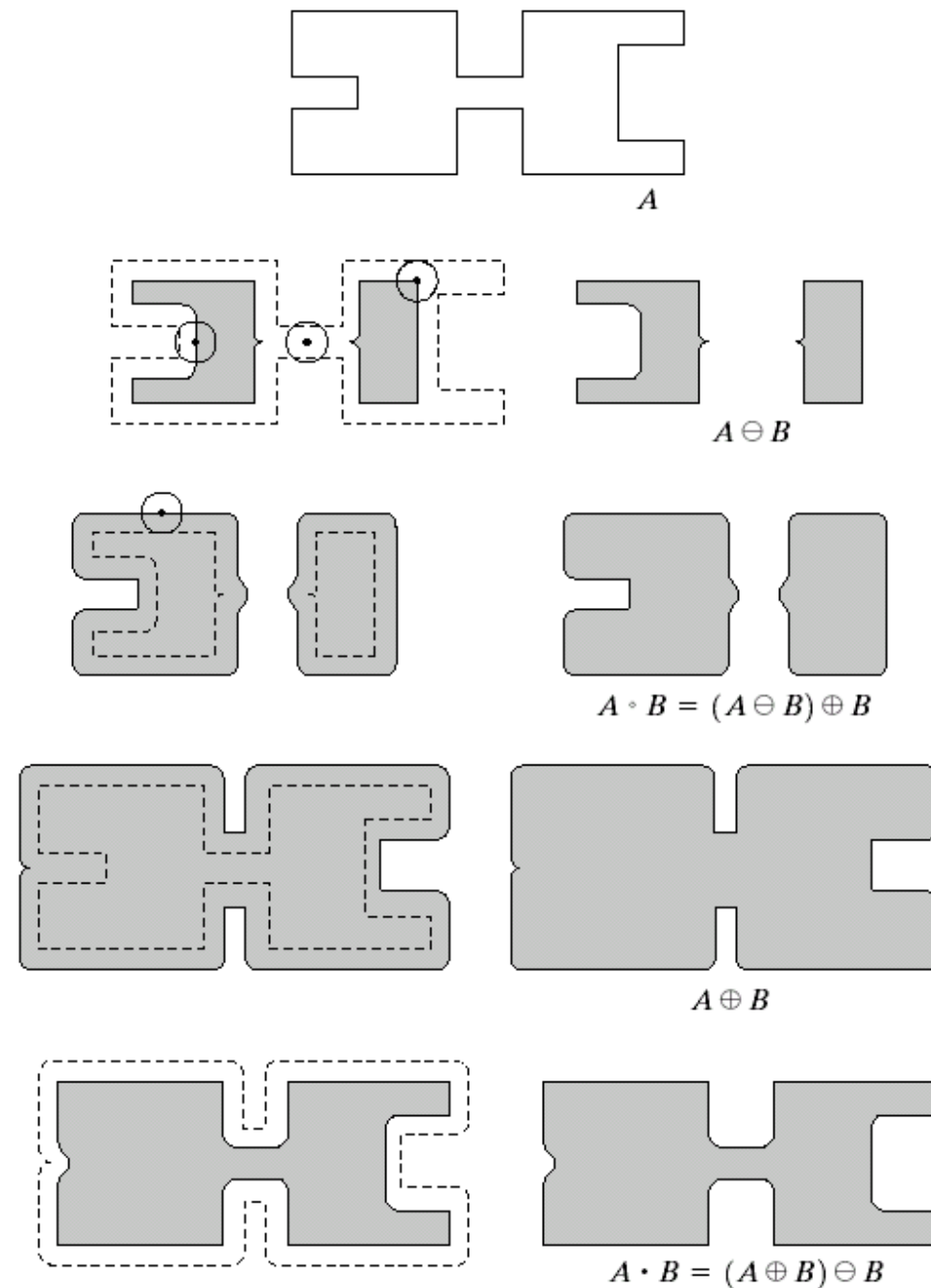


FIGURE 9.9 (a) Structuring element B “rolling” on the outer boundary of set A . (b) Heavy line is the outer boundary of the closing. (c) Complete closing (shaded).

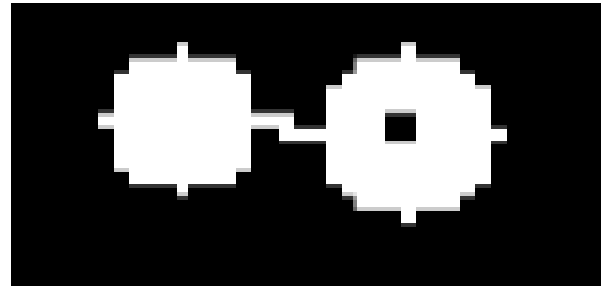
$$A \bullet B = (A \oplus B) \ominus B$$

a
b c
d e
f g
h i

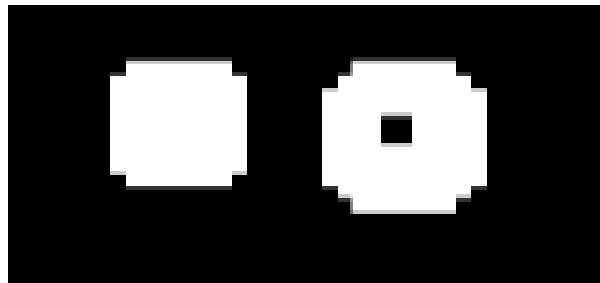
FIGURE 9.10
Morphological opening and closing. The structuring element is the small circle shown in various positions in (b). The dark dot is the center of the structuring element.



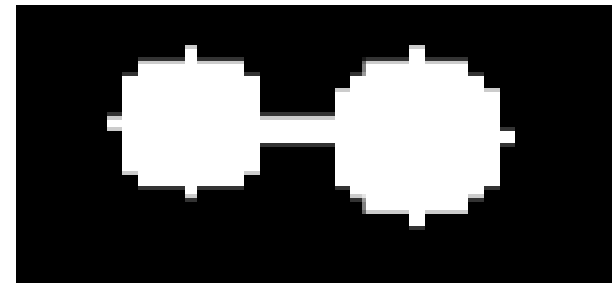
Useful: open & close



A



opening of A
→ removal of small protrusions, thin
connections, ...

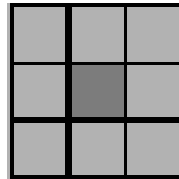


closing of A
→ removal of holes

Note: repeated openings/closings has no effect!

Application: filtering

Application:
filtering



1. erode
 $A \ominus B$



2. dilate
 $(A \ominus B) \oplus B = A \circ B$



3. dilate
 $(A \ominus B) \oplus B$



4. erode
 $((A \ominus B) \oplus B) \ominus B = (A \circ B) \ominus B$

Morphological Algorithms

Using the simple technique we have looked at so far we can begin to consider some more interesting morphological algorithms

We will look at:

- Boundary extraction
- Region filling

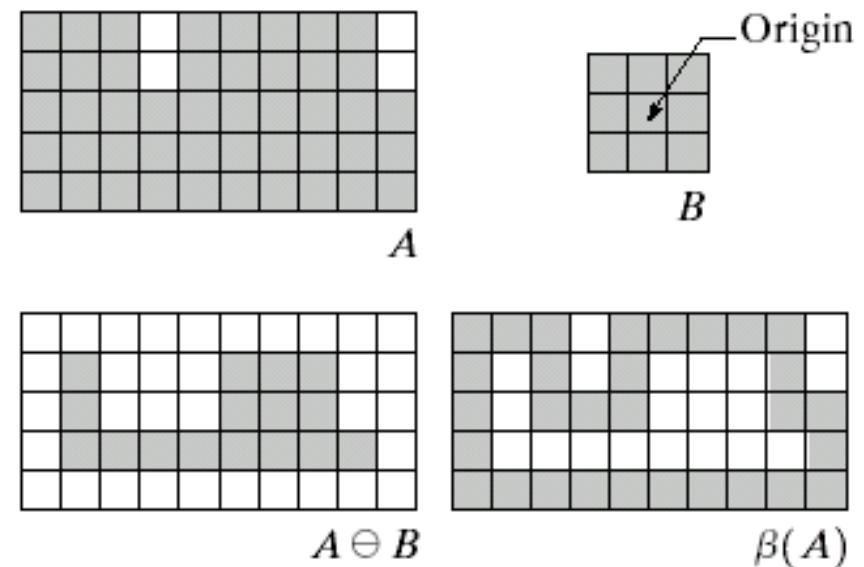
There are lots of others as well though:

- Extraction of connected components
- Thinning/thickening
- Skeletonisation

Boundary Extraction

a	b
c	d

FIGURE 9.13 (a) Set A . (b) Structuring element B . (c) A eroded by B . (d) Boundary, given by the set difference between A and its erosion.



$$\beta(A) = A - (A \ominus B)$$

Example



a b

FIGURE 9.14

(a) A simple binary image, with 1's represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

Region Filling

$$X_k = (X_{k-1} \oplus B) \cap A^c \quad k = 1, 2, 3, \dots$$

a	b	c
d	e	f
g	h	i

FIGURE 9.15

Region filling.

(a) Set A .

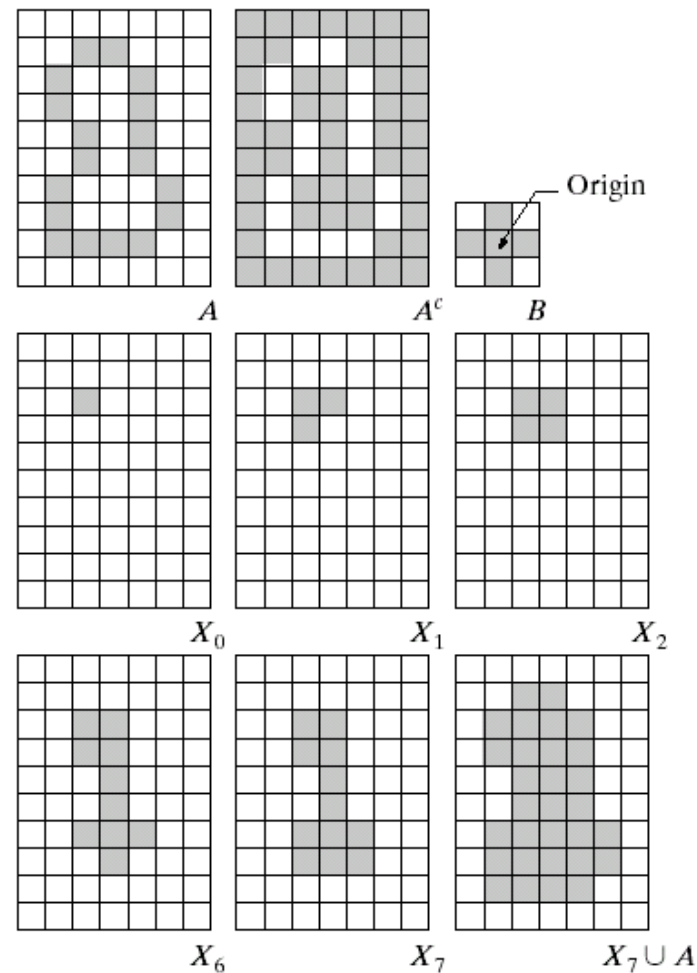
(b) Complement of A .

(c) Structuring element B .

(d) Initial point inside the boundary.

(e)–(h) Various steps of Eq. (9.5-2).

(i) Final result [union of (a) and (h)].



Example

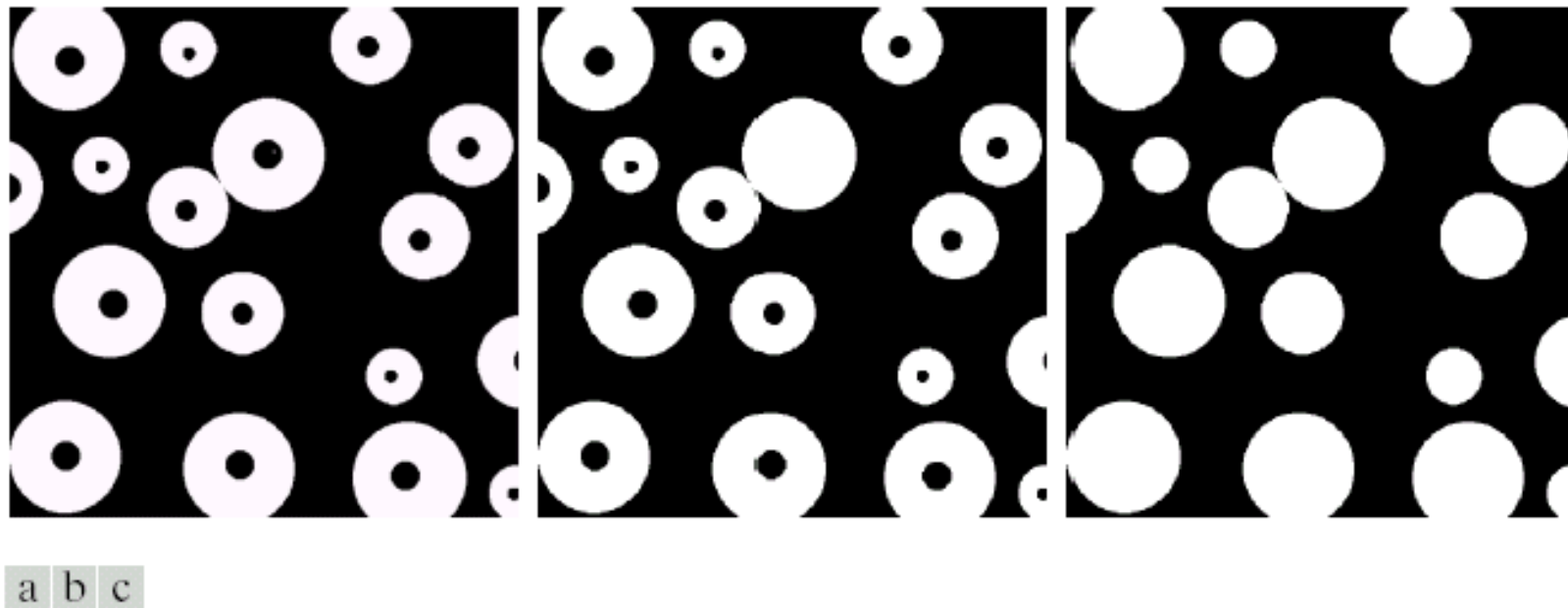


FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the region-filling algorithm). (b) Result of filling that region (c) Result of filling all regions.

Extraction of connected components

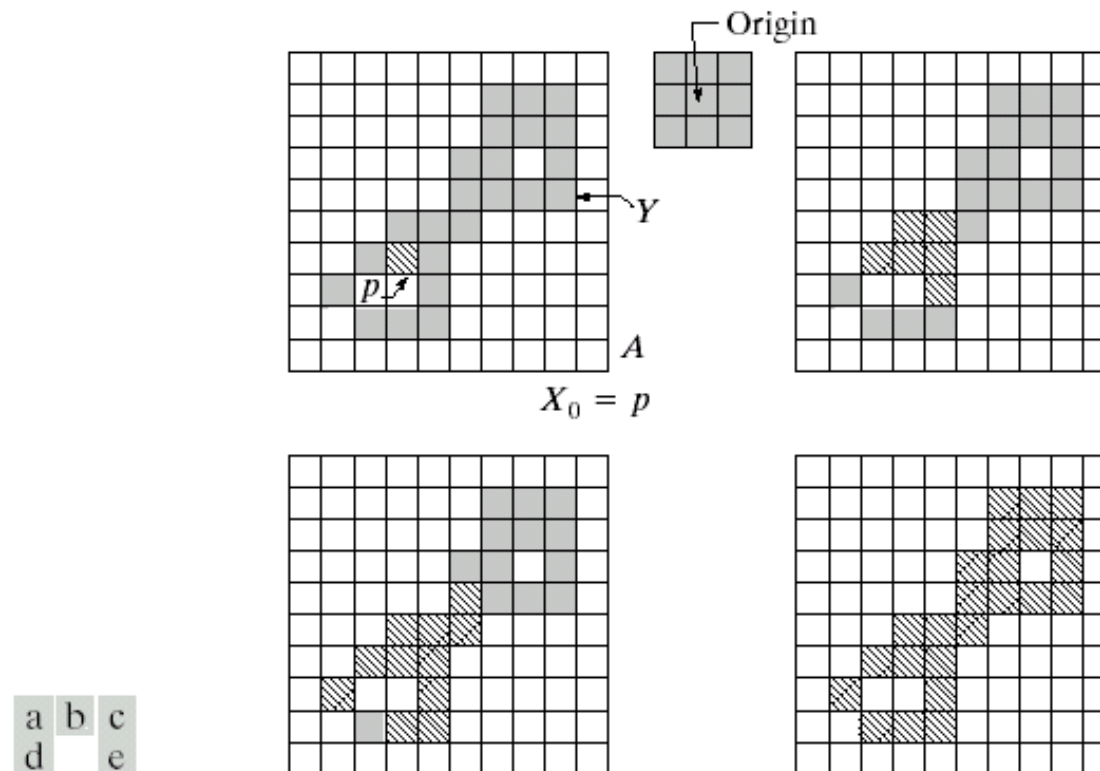


FIGURE 9.17 (a) Set A showing initial point p (all shaded points are valued 1, but are shown different from p to indicate that they have not yet been found by the algorithm). (b) Structuring element. (c) Result of first iterative step. (d) Result of second step. (e) Final result.

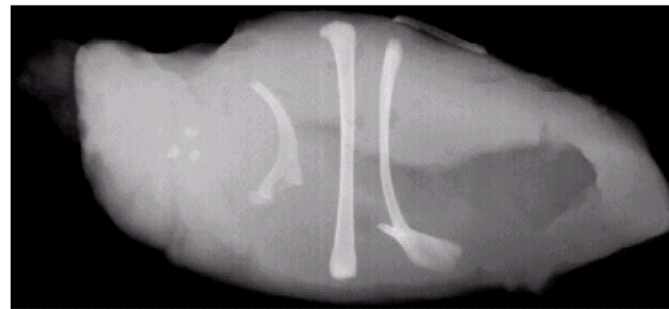
$$X_k = (X_{k-1} \oplus B) \cap A \quad k = 1, 2, 3, \dots$$

Example

a
b
c d

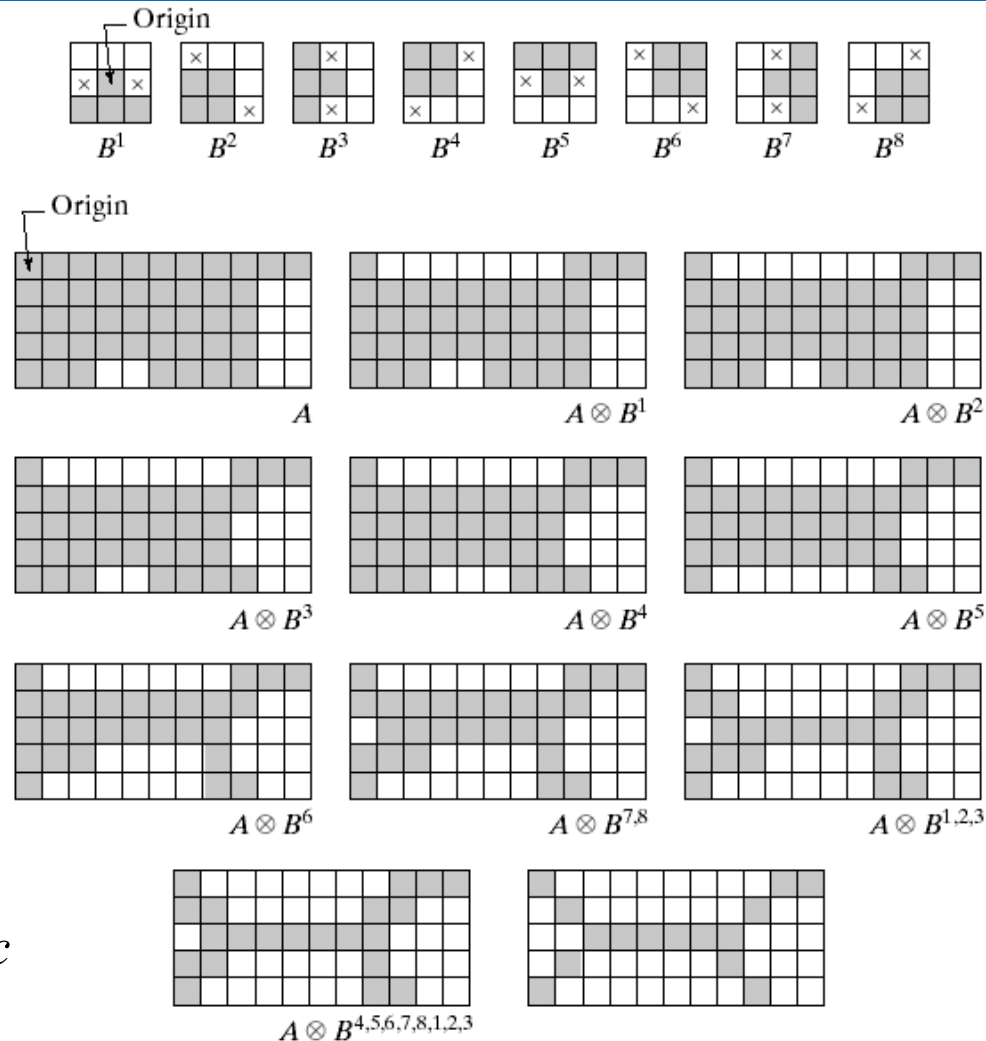
FIGURE 9.18

(a) X-ray image of chicken filet with bone fragments.
(b) Thresholded image. (c) Image eroded with a 5×5 structuring element of 1's.
(d) Number of pixels in the connected components of (c). (Image courtesy of NTB Elektronische Geraete GmbH, Diepholz, Germany, www.ntbxray.com.)



Connected component	No. of pixels in connected comp
01	11
02	9
03	9
04	39
05	133
06	1
07	1
08	743
09	7
10	11
11	11
12	9
13	9
14	674
15	85

Thinning



$$A \otimes B = A - (A * B)$$

$$= A \cap (A * B)^c$$

a
b c d
e f g
h i j
k l

FIGURE 9.21 (a) Sequence of rotated structuring elements used for thinning. (b) Set A . (c) Result of thinning with the first element. (d)–(i) Results of thinning with the next seven elements (there was no change between the seventh and eighth elements). (j) Result of using the first element again (there were no changes for the next two elements). (k) Result after convergence. (l) Conversion to m -connectivity.

Thickening

$$A \odot B = A \cup (A * B)$$

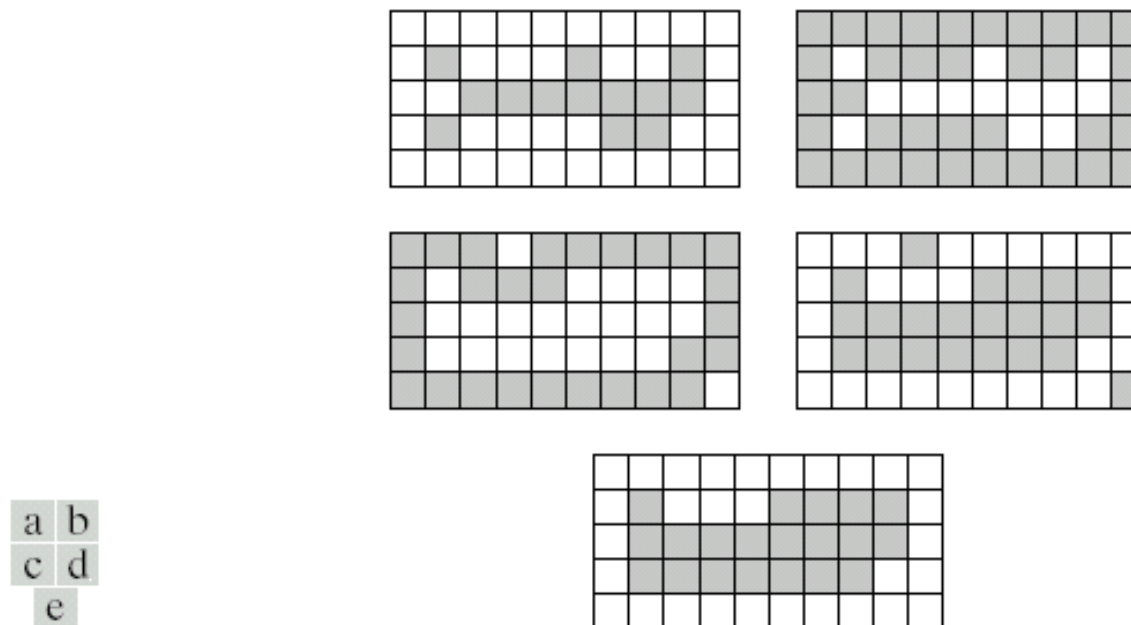


FIGURE 9.22 (a) Set A . (b) Complement of A . (c) Result of thinning the complement of A . (d) Thickened set obtained by complementing (c). (e) Final result, with no disconnected points.

Skeletons

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

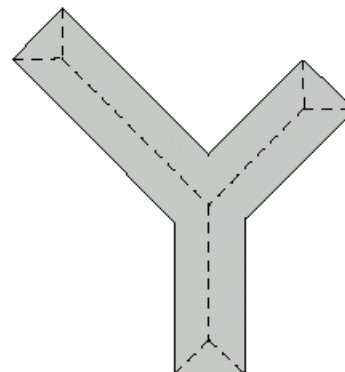
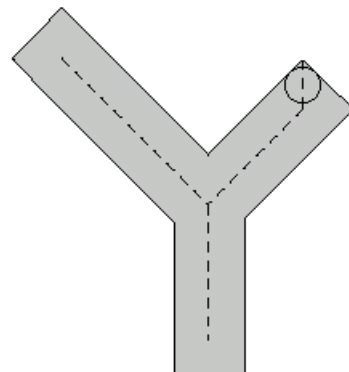
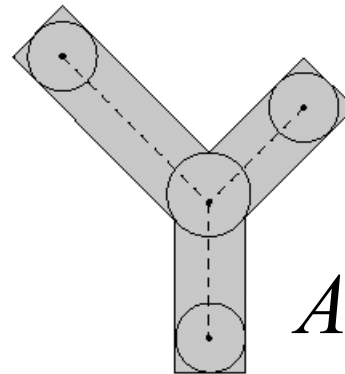
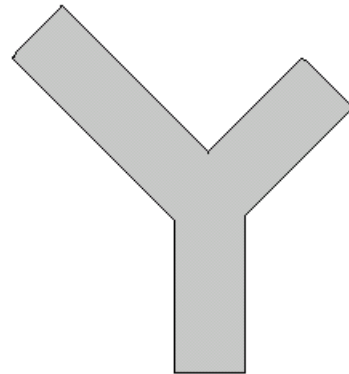
$$S_k(A) = (A \ominus kB) - (A \ominus kB) \circ B$$

$$K = \max\{k \mid (A - kB) \neq \Phi\}$$

a b
c d

FIGURE 9.23

(a) Set A .
(b) Various positions of maximum disks with centers on the skeleton of A .
(c) Another maximum disk on a different segment of the skeleton of A .
(d) Complete skeleton.



$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

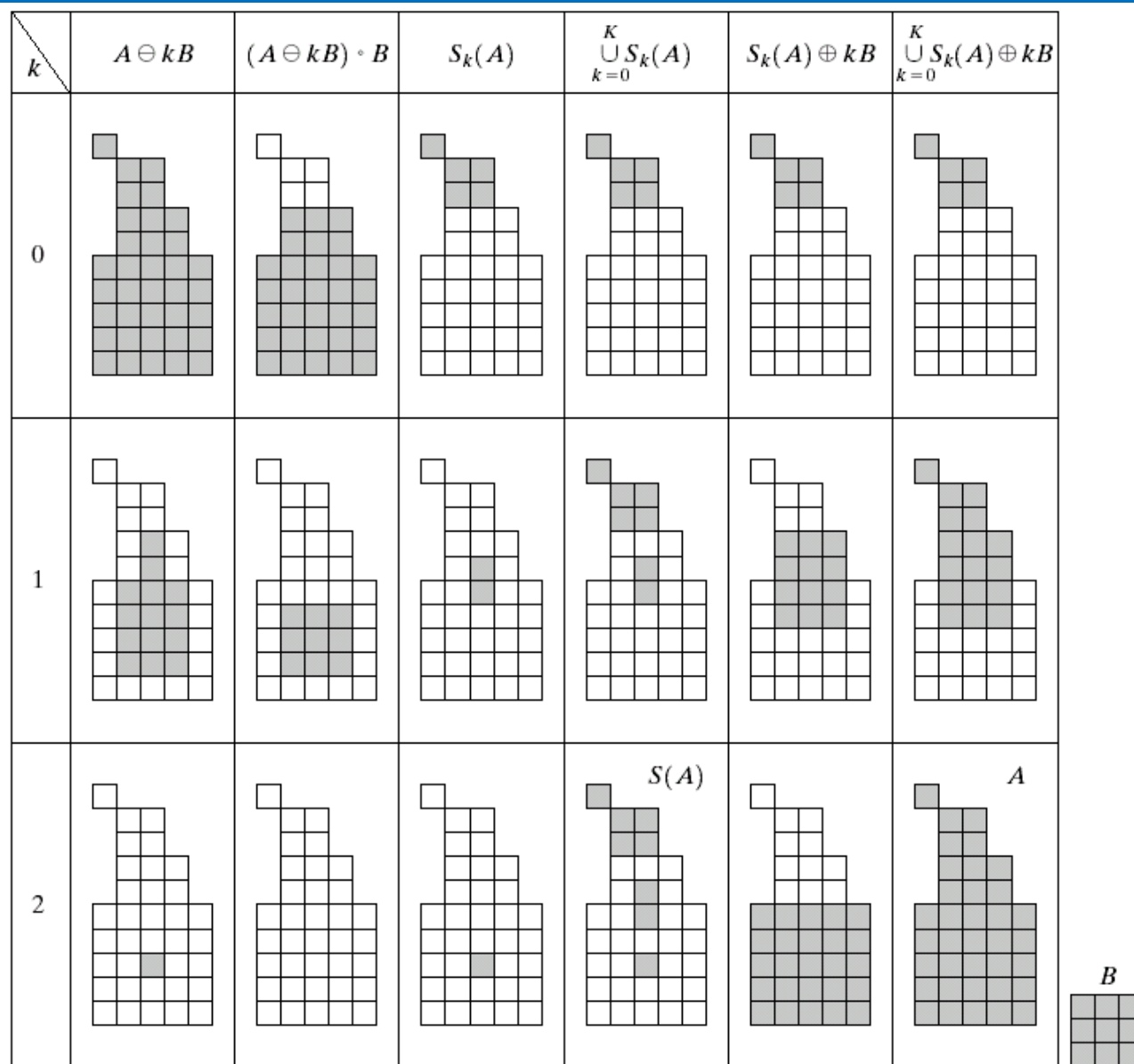


FIGURE 9.24 Implementation of Eqs. (9.5-11) through (9.5-15). The original set is at the top left, and its morphological skeleton is at the bottom of the fourth column. The reconstructed set is at the bottom of the sixth column.

TABLE 9.2

Summary of morphological operations and their properties.

Operation		Equation	Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Translation		$(A)_z = \{w w = a + z, \text{ for } a \in A\}$	Translates the origin of A to point z .
Reflection		$\hat{B} = \{w w = -b, \text{ for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement		$A^c = \{w w \notin A\}$	Set of points not in A .
Difference		$A - B = \{w w \in A, w \notin B\}$ $= A \cap B^c$	Set of points that belong to A but not to B .
Dilation		$A \oplus B = \{z (\hat{B})_z \cap A \neq \emptyset\}$	“Expands” the boundary of A . (I)
Erosion		$A \ominus B = \{z (B)_z \subseteq A\}$	“Contracts” the boundary of A . (I)
Opening		$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)
Closing		$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)

Hit-or-miss transform	$A \circledast B = (A \ominus B_1) \cap (A^c \ominus B_2)$ $= (A \ominus B_1) - (A \oplus \hat{B}_2)$	The set of points (coordinates) at which, simultaneously, B_1 found a match (“hit”) in A and B_2 found a match in A^c .
Boundary extraction	$\beta(A) = A - (A \ominus B)$	Set of points on the boundary of set A . (I)
Region filling	$X_k = (X_{k-1} \oplus B) \cap A^c; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Fills a region in A , given a point p in the region. (II)
Connected components	$X_k = (X_{k-1} \oplus B) \cap A; X_0 = p \text{ and } k = 1, 2, 3, \dots$	Finds a connected component Y in A , given a point p in Y . (I)
Convex hull	$X_k^i = (X_{k-1}^i \circledast B^i) \cup A; i = 1, 2, 3, 4;$ $k = 1, 2, 3, \dots; X_0^i = A; \text{ and}$ $D^i = X_{\text{conv}}^i.$	Finds the convex hull $C(A)$ of set A , where “conv” indicates convergence in the sense that $X_k^i = X_{k-1}^i$. (III)

Operation	Equation	Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Thinning	$A \otimes B = A - (A \circledast B)$ $= A \cap (A \circledast B)^c$ $A \otimes \{B\} =$ $((\dots ((A \otimes B^1) \otimes B^2) \dots) \otimes B^n)$ $\{B\} = \{B^1, B^2, B^3, \dots, B^n\}$	Thins set A . The first two equations give the basic definition of thinning. The last two equations denote thinning by a sequence of structuring elements. This method is normally used in practice. (IV)
Thickening	$A \odot B = A \cup (A \circledast B)$ $A \odot \{B\} =$ $((\dots (A \odot B^1) \odot B^2 \dots) \odot B^n)$	Thickens set A . (See preceding comments on sequences of structuring elements.) Uses IV with 0's and 1's reversed.

TABLE 9.2
Summary of morphological results and their properties.
(continued)

Skeletons

$$S(A) = \bigcup_{k=0}^K S_k(A)$$

$$S_k(A) = \bigcup_{k=0}^K \{ (A \ominus kB) - [(A \ominus kB) \circ B] \}$$

Reconstruction of A :

$$A = \bigcup_{k=0}^K (S_k(A) \oplus kB)$$

Finds the skeleton $S(A)$ of set A . The last equation indicates that A can be reconstructed from its skeleton subsets $S_k(A)$. In all three equations, K is the value of the iterative step after which the set A erodes to the empty set. The notation $(A \ominus kB)$ denotes the k th iteration of successive erosion of A by B . (I)

Pruning

$$X_1 = A \otimes \{B\}$$

$$X_2 = \bigcup_{k=1}^8 (X_1 \circledast B^k)$$

$$X_3 = (X_2 \oplus H) \cap A$$

$$X_4 = X_1 \cup X_3$$

X_4 is the result of pruning set A . The number of times that the first equation is applied to obtain X_1 must be specified. Structuring elements V are used for the first two equations. In the third equation H denotes structuring element I .

Summary

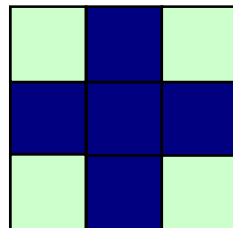
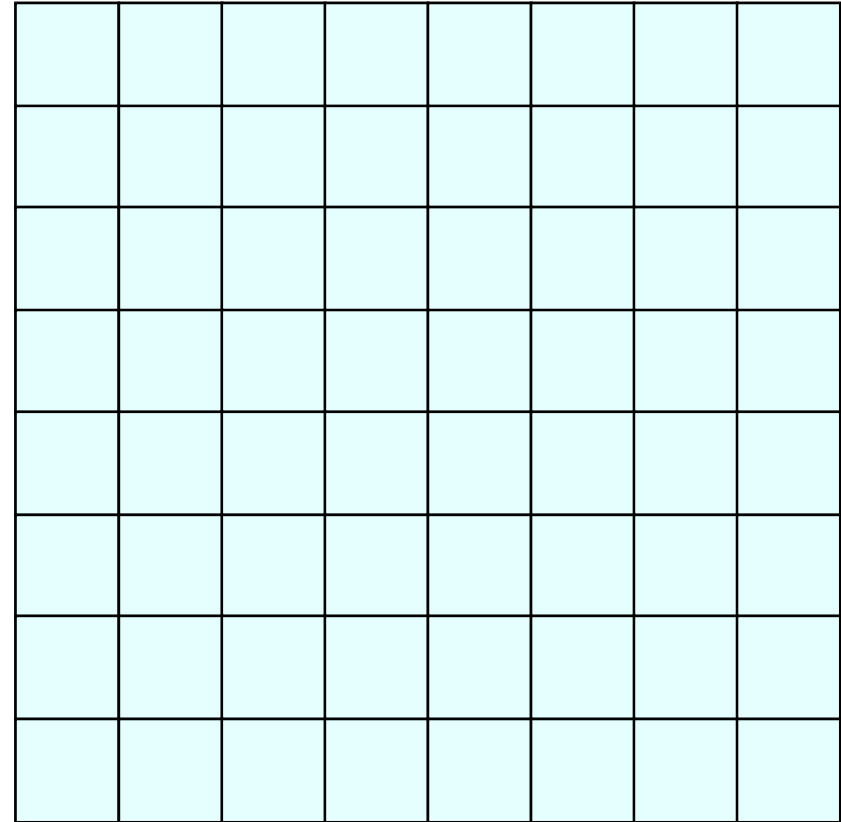
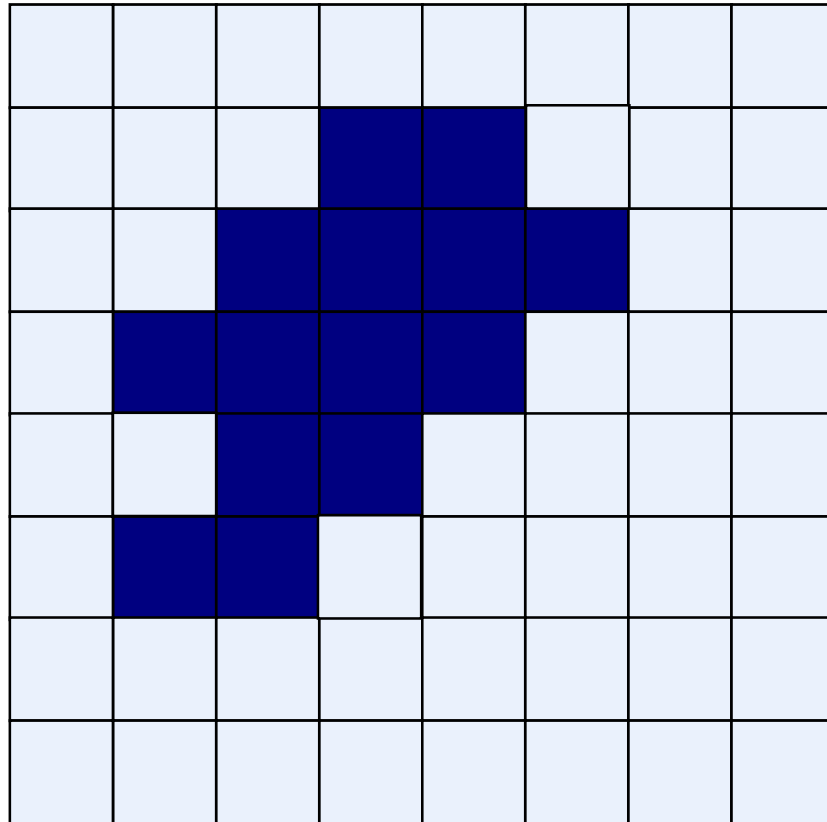
The purpose of morphological processing is primarily to remove imperfections added during segmentation

The basic operations are *erosion* and *dilation*

Using the basic operations we can perform *opening* and *closing*

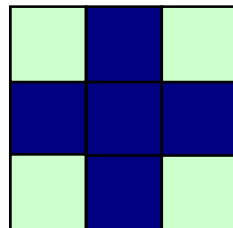
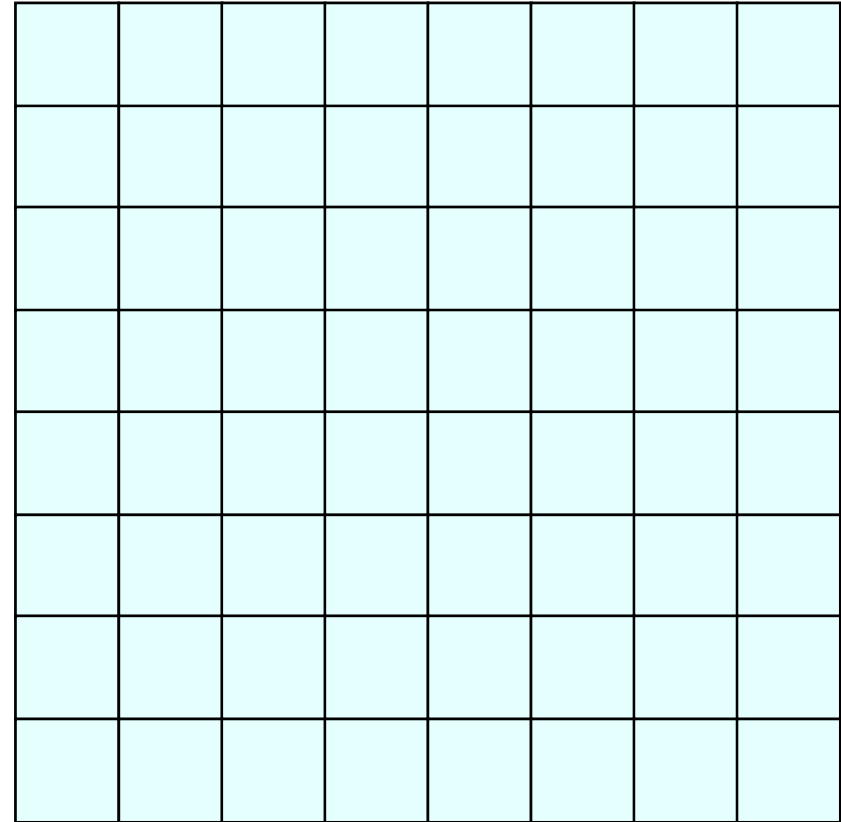
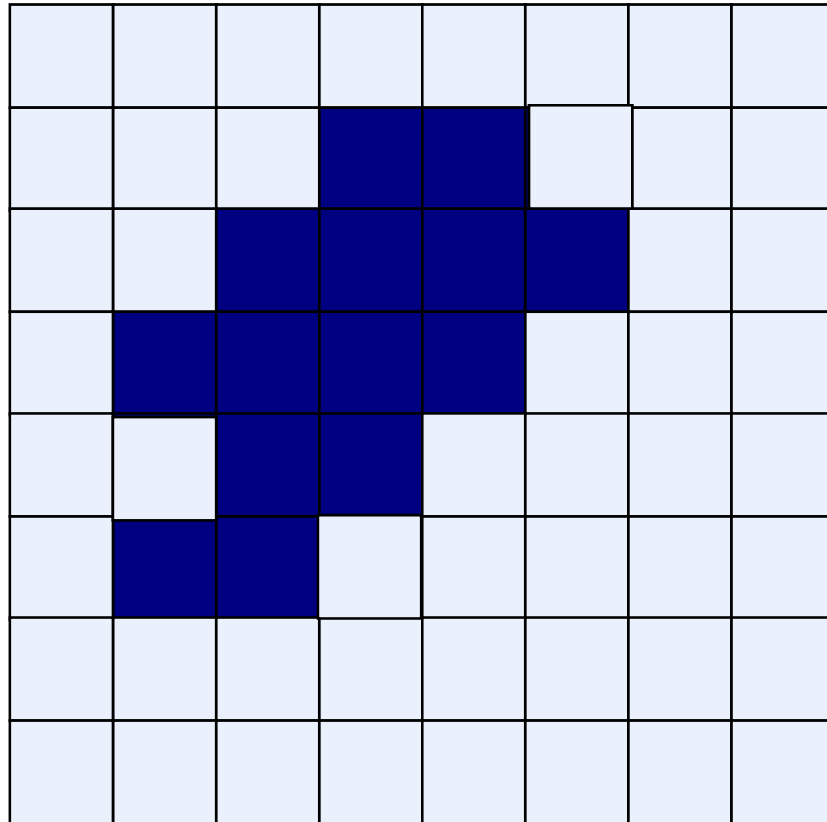
More advanced morphological operation can then be implemented using combinations of all of these

Erosion Example



Structuring Element

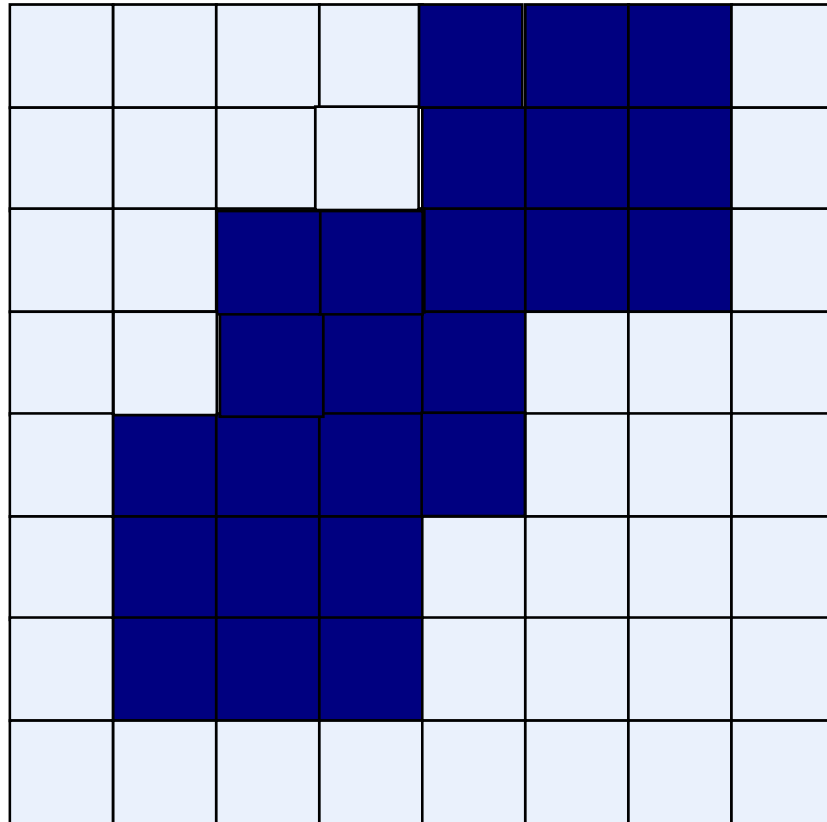
Dilation Example



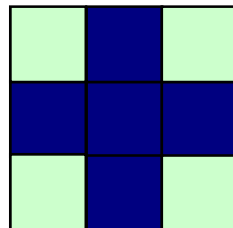
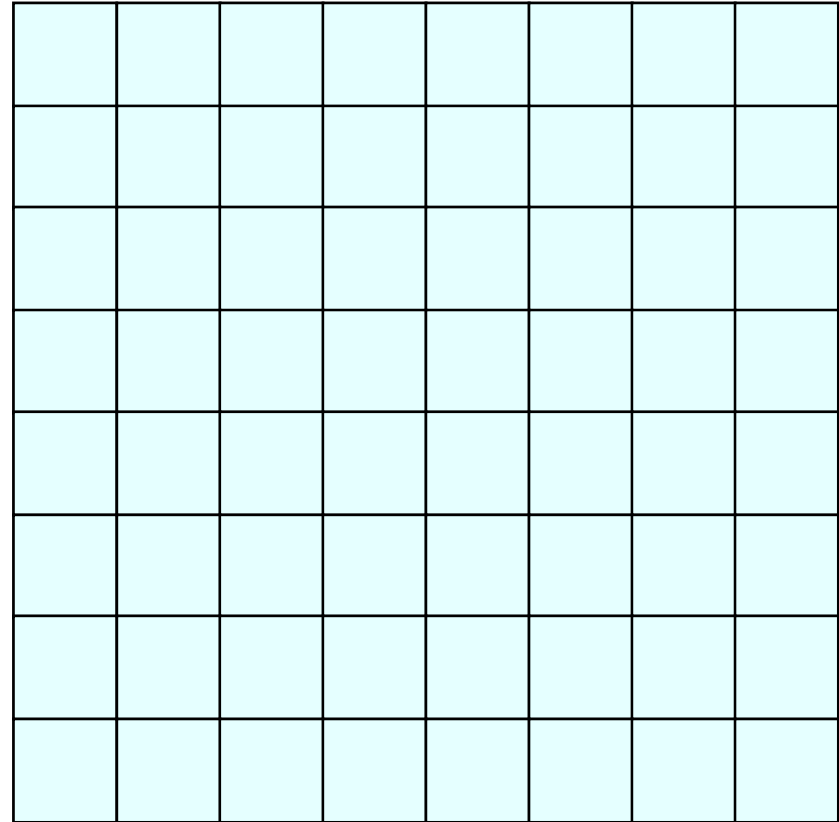
Structuring Element

Opening Example

Original Image



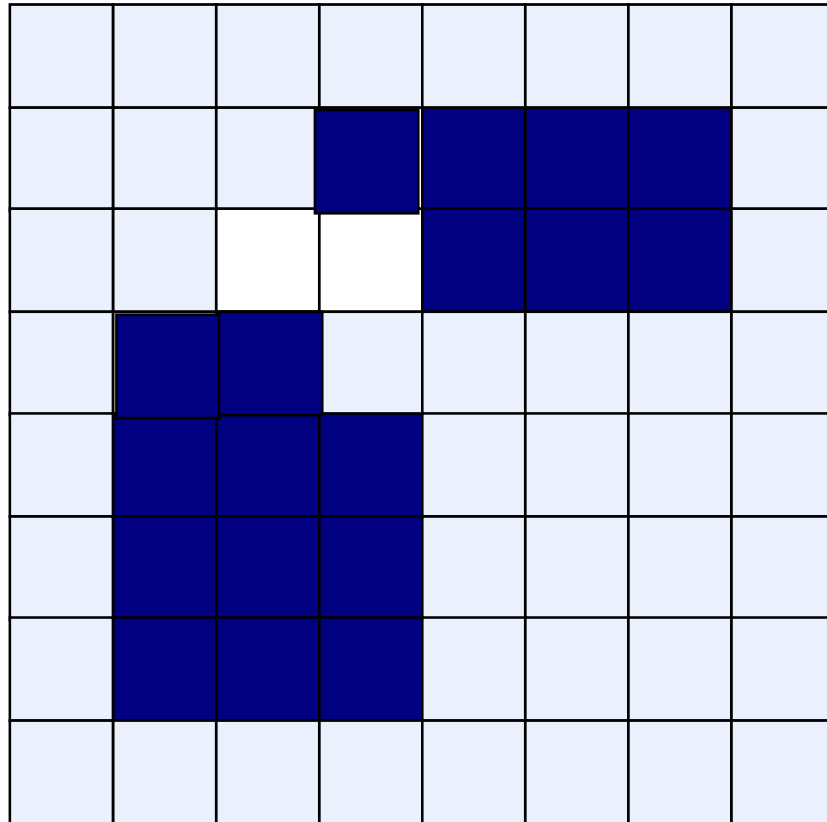
Processed Image



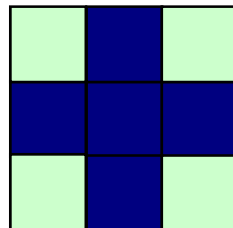
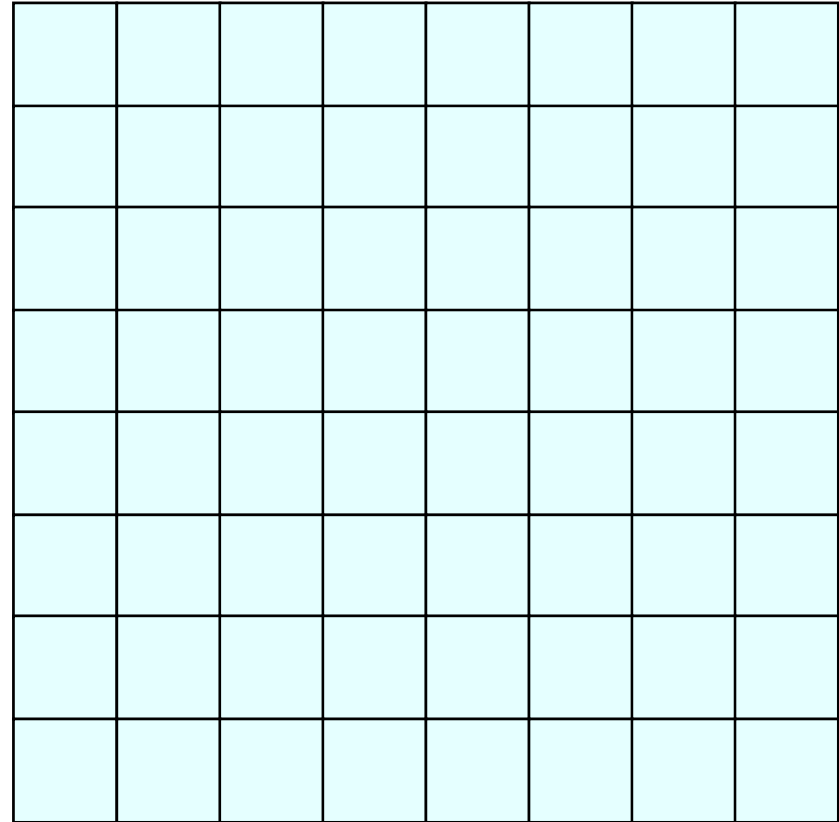
Structuring Element

Closing Example

Original Image



Processed Image



Structuring Element

